DUBAI RAPID LINK CONSORTIUM

DUBAI METRO PROJECT OFFICE

Contract No.: DM001	CDRL No.:
Project Title: DUBAI METRO	CDRL Title:

Document Title:

Red Line - Rashidiya Main Depot
General Overhaul Workshop & Lathe on Pit
Supplementary Structural Design Calculations
Building 03
DCP 4 Submission

Revision History

A1	04-Jun-09	First Issue	Signed below	Signed below
MARK	**DATE**	**DESCRIPTION**	**RAIL**	**CIVIL**

APPROVED	Project Director	Yoji Hiraoka	Y. Hiraoka

		Checked By (Engineering Director)	T.Shirasuna
		Checked By (QA/QC Manager)	J. Whorlow
Checked By	N/A	Checked By (Project Manager)	N/A
Checked By	N/A	Checked By (Engineering Manager)	M. Makino
Prepared By	N/A	Issued By	B. Maney
DATE	-	DATE	04 June 2009
RAIL SYSTEM		**CIVIL JV**	

CONTRACTOR'S DOCUMENT No.:	DOCUMENT No.: **DM001-E-MAD-WGO-DR-DCC-371922**	REVISION **A1**

DUBAI RAPID LINK CONSORTIUM

Dubai Metro, UAE
Red Line Main Depot - Rashidiya
General Overhaul Workshop & Lathe on Pit
(Building 3)

Supplementary Structural Design Calculations
(DCP4 Re-Submission)

May 2009

ATKINS
WS Atkins & Partners Overseas

Client: **JT Metro JV**	***Contract No.*** *(if any):*
Project Title: **Dubai Metro, UAE**	***Project No.:*** **JRD4561**
Document No.: **DM001-E-MAD-WGO-DR-DCC-371922 A1**	***Controlled Copy No.:***
Document Title: **Red Line Main Depot - Rashidya General Overhaul Workshop & Lathe on Pit (Building 3) Supplementary Structural Design Calculations (DCP4 Re-Submission)**	
Covering Letter / Transmittal Ref. No.:	***Date of Issue:*** **12 May 2009**

Revision, Review and Approval/Authorization Records

A1	DCP4 Submission	C. Camba May 09	D. Lawale May 09	B. Maney May 09
Revision	Description	Prepared by / date	Reviewed by / date	App. or Auth. by / date

Distribution (if insufficient space, please use separate paper)

Controlled Copy No.	Issued to
1	JT Metro JV
2	Atkins (Office Copy)
3	Atkins (Drake Lawale)
4	Atkins (Cid Camba)

Note: App. and Auth. mean "Approved" and "Authorized" respectively.

Table of Contents

1 Introduction

The design report is submitted as a supplementary calculation for Paint Booth Area (revised) and AHU Platforms in Lath-on-pit area etc.

Please refer to previously submitted Documents no. DM001-E-MAD-CDP-DR-DCC-371942-Rev-A5 (Building) & DM001-E-MAD-CDP-DR-DCC-371920-A1(AHU Platforms) for the design calculations of all other structural members, which are not covered in this document.

PAINT BOOTH AREA DESIGN

The structural design and details of the Paint Booth Area has been updated to cover the changes on the support frame's member size, footprint dimension and total vertical height. Loadings have been adjusted accordingly and the frame has been designed using Staad's Steel Design facility under the BS5950 code.

The footing has been designed considering the adjoining footing at grid 3-A/3-13 Dwg. DM001-E-MAD-WGO-DD-DCC-370701-D1 which was already built. Stability checks has been performed considering soil-structure interaction of the integrated model of foundation and superstructure. Concrete Design has been performed using Staad's element design facility under the BS8110 code.

AHU PLATFORM DESIGN (Lath On Pit)

Due to the changes of enhanced air-conditioning requirements in Depot buildings and increase in weight of AHU equipments, this supplementary calculation supports the proposal of mounting the AHU on raised platform that across Lath on Pit area.

The design consideration is based on the followings:

- The necessary investigation/evaluation on the AHU structural support Methods and AHU installation/maintenance system.
- Latest AHU equipment information.
- Site constraints.

The proposed structural system is to use a column supporting steel platform floors to take all the gravity loads (including the AHU equipment units). The steel frames are designed with pinned connection at support location, and the integrated bracing system gives the stability for the structure in resisting the lateral forces. In addition, the steel plate at platform level will give additional restraints to the frames. The proposed structural system will be able to simplify the connections between the beams and columns and increase the efficiency of the columns. The design is append on page 48.

2. Paint Booth Support Frame and Foundation

ATKINS A member of the WS Atkins Group	Project RL RASHIDIYA DEPOT			Job ref.	
	Part of Structure PAINT BOOTH AREA			Calc. sheet no. /	rev. /
PO Box 5620 Dubai, United Arab Emirates	Drawing ref.	Calc. by COC	Date 12 MAY '09	Check by	Date

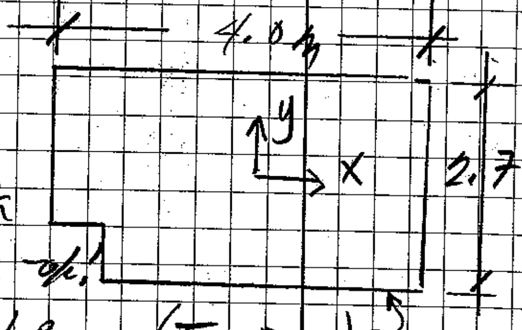
Ref.	Calculations	Output

STRUCTURAL DESIGN OF PAINTH BOOTH SUPPORT FRAME AND FOUNDATION (Bldg. 3)

Loadings: (Refer to pages 7-10)

Support frame: (Refer to pages 32-36)
∴ Use 200X200X6 SHS colums and 125×125×4.5 mm SHS horizontal and diagonal bracings.

Foundation design:
- check for stability; (from p. 44)



Max soil pressure = 42 kPa
< 150 kPa - ok!

Min soil pressure = + 0.04 kPa (T = 750)
no uplift - ok!

- Solving for reinforcement;
As req'd = 975 mm²/m (p. 37 and 38 of staad calcs)
As provided = 1340 mm²/m > 975
(T16-150) - ok! (6)

Loadings :

SDL :

Duct s/w + accessories /support, $W_1 = 0.45$ kN/m

For 1.5m length, $W_1 = 0.45 \times 1.5$
$= 0.7$ kN

Assume 2 number of point loads (0.35 kN each) acting at 1.5m ht. interval.

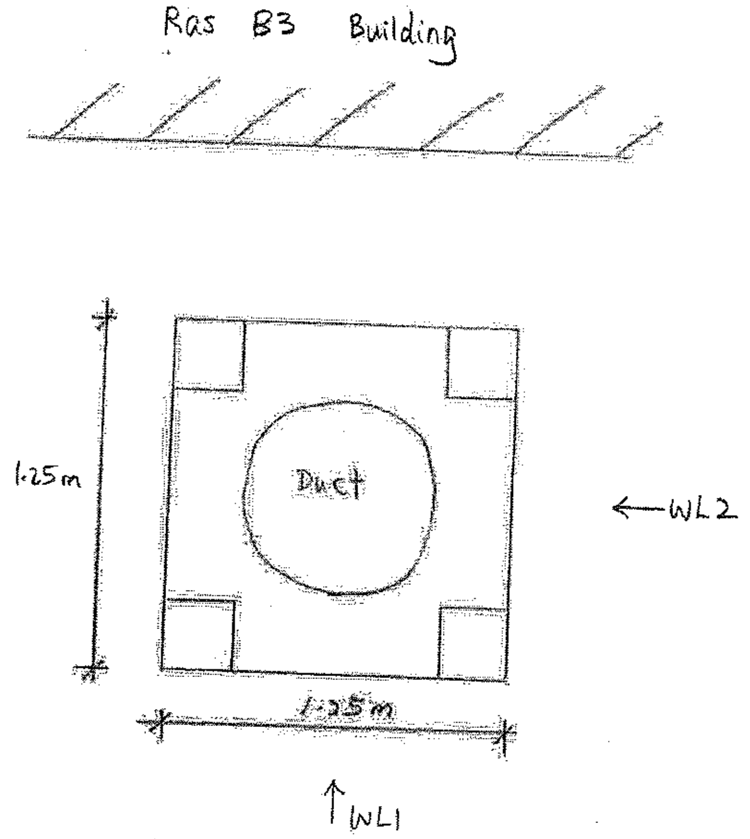
LL :

Assume 2 persons for maintenance which is equivalent to 2 number of point loads (1 kN each) acting on top level. — Not used, negligible

Calculation/Sketch

Site Ras B3 | Date | Job No.
Designed by | Sheet No.

WL:



PLAN

$D = 1.25m$
$H = 15\ m$
$D/H = 0.08 < 1$

$WL1 = q \times [Cpe\ (front) - Cpe\ (rear)]$
$= 1.24 \times [0.85 - (0)]$
$= 1.05\ kPa$

$WL2 = 1.24 \times [0.85 - (-0.5)]$
$= 1.67\ kPa$

Refer to BS 6399 (see attached)

Note:
For WL1,
Cpe (rear) = 0
because the wind pressure on the leeward side is obstructed by Ras B3 building wall.

2.4.1.5 The values in Table 5 are also valid for non-vertical walls within ±15° of the vertical. Values outside this range should be obtained from 3.3.1.4.

2.4.2 Polygonal buildings

External pressure coefficients for the vertical walls of buildings with corner angles other than 90° should be obtained using the procedures set out in 3.3.1.2.

Overall forces may be calculated using the pressure coefficients of Table 5 together with equation 23 of 3.1.3.3.2.

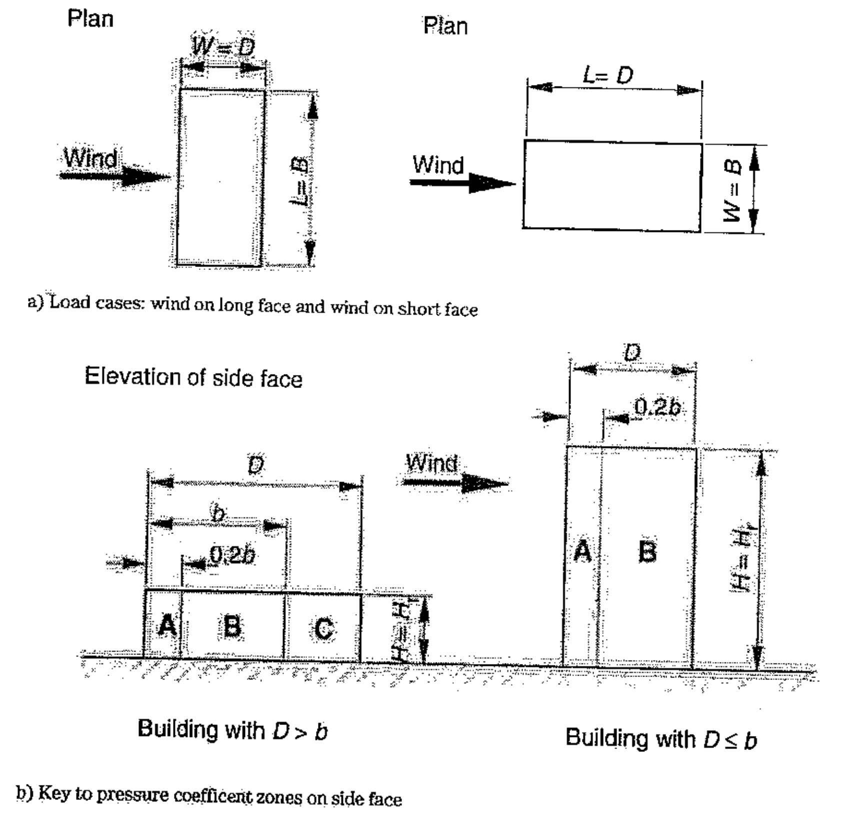


Figure 12 — Key to wall pressure data

Table 5 — External pressure coefficients C_{pe} for vertical walls

Vertical wall face	Span ratio of building		Vertical wall face		Exposure case	
	$D/H \le 1$	$D/H \ge 4$			Isolated	Funnelling
Windward (front)	+0.85	+0.6	Side	Zone A	–1.3	–1.6
Leeward (rear)	–0.5	–0.5		Zone B	–0.8	–0.9
				Zone C	–0.5	–0.9

NOTE Interpolation may be used in the range $1 < D/H < 4$. See 2.4.1.4 for interpolation between isolated and funnelling.

Site:	Ras 83 Paint Booth Exhaust Duct Support Frame Earthquake Loading	Date:	15-Jan-09	Job No.:	68010.21
		Designed by:	OEH	Sheet No.:	

Esitmation of Earthquake Loading

Total Dead Load on the Building Column   W = Given below   (including part of Live Load)
Height of the Column h_n = Given below

Design Information:

1) Seismic Zone = 2A   (from Design Brief)
2) Occupancy Category = 1   (Tanle- 16-K, Conservatively adopt 1)
3) Lateral Force procedure = **Static Procedure**   (Section 1629.8 seismic zone 2A, structures not more than 5 stories)

Earthquake Loads

4) **Base Shear** $V_{base\text{-}S} = (C_V \times I \times W) / (R \times T)$ (Section 1630.2.1)
Seismic Coefficient C_V = 0.32   (Table 16-R assuming soil profile type S_D)
Importance Factor I = 1.00   (Table 16-K)
Numeric. Coef. Strength & Ductility R = 3.50   (Table 16-N for OMRT concrete only)
Elastic Fundamental Period of Vibration $T = C_t\ (h_n)^{3/4}$ Sec.
Numerical Coefficient C_t = 0.0731   (Concrete frame t.b.c)

5) **Limiting Values of Base shear** $V_{base\text{-}S\text{-}Max} = (2.5 \times C_a \times I \times W) / R$ = 1.56
$V_{base\text{-}S\text{-}Min} = 0.11 \times C_a \times I \times W$ = 0.24

Design Earthquake Loads

(Section 1630.6)

$E_x = \rho \times \text{Min.} (E_h \text{ or } V_{base\text{-}S\text{-}Max}) + E_V$

$E_{x\text{-}max} = \Omega \times \text{Min.} (E_h \text{ or } V_{base\text{-}S\text{-}Max})$

$V_{base\text{-}S} = E_h$ = Earthquake load in the element due to the base shear, V, as set out above (for non-structural components refer to section 1632)

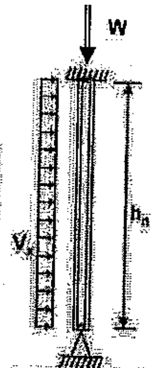
Vert. component of EQ ground motion $E_v = 0.5\,C_a\,I\,D$ (Table 16-K)
Seismic Coefficient C_a = 0.22   (Table 16-Q, for soil type S_D)
Seismic Force Amplification Factor Ω = 2.80   (Table 16-N)
Dead Load on the element   W = D = Given below
$\rho = 2 - 6.1 / (r_{max} \sqrt{A_B})$ = 1.25 (conservative) $(1.0 \le \rho \le 1.5)$
r_{max} = The maximum element-storey shear ratio
A_B = Ground floor area of the structure

Distribution of Horizontal Forces:

Distribution of Horizontal Force on the Column V_x = E_x / h_n

Col	G/L		Load, W, kN	Ht h_n, m	Time, T sec.	$V_{base\text{-}S}$ kN	$V_{base\text{-}S}$ kN	E_v kN	E_x kN	$E_{x\text{-}max}$ kN	V_x kN/m	$V_{x\text{-}max}$ kN/m
1001			9.9	15.00	0.56	1.6	1.56	1.089	3.0	4.4	0.20	0.3
1002			9.6	15.00	0.56	1.6	1.51	1.056	2.9	4.2	0.20	0.3
1003			10.0	15.00	0.56	1.6	1.57	1.1	3.1	4.4	0.20	0.3
1004			9.7	15.00	0.56	1.6	1.52	1.067	3.0	4.3	0.20	0.3

From LC 2002



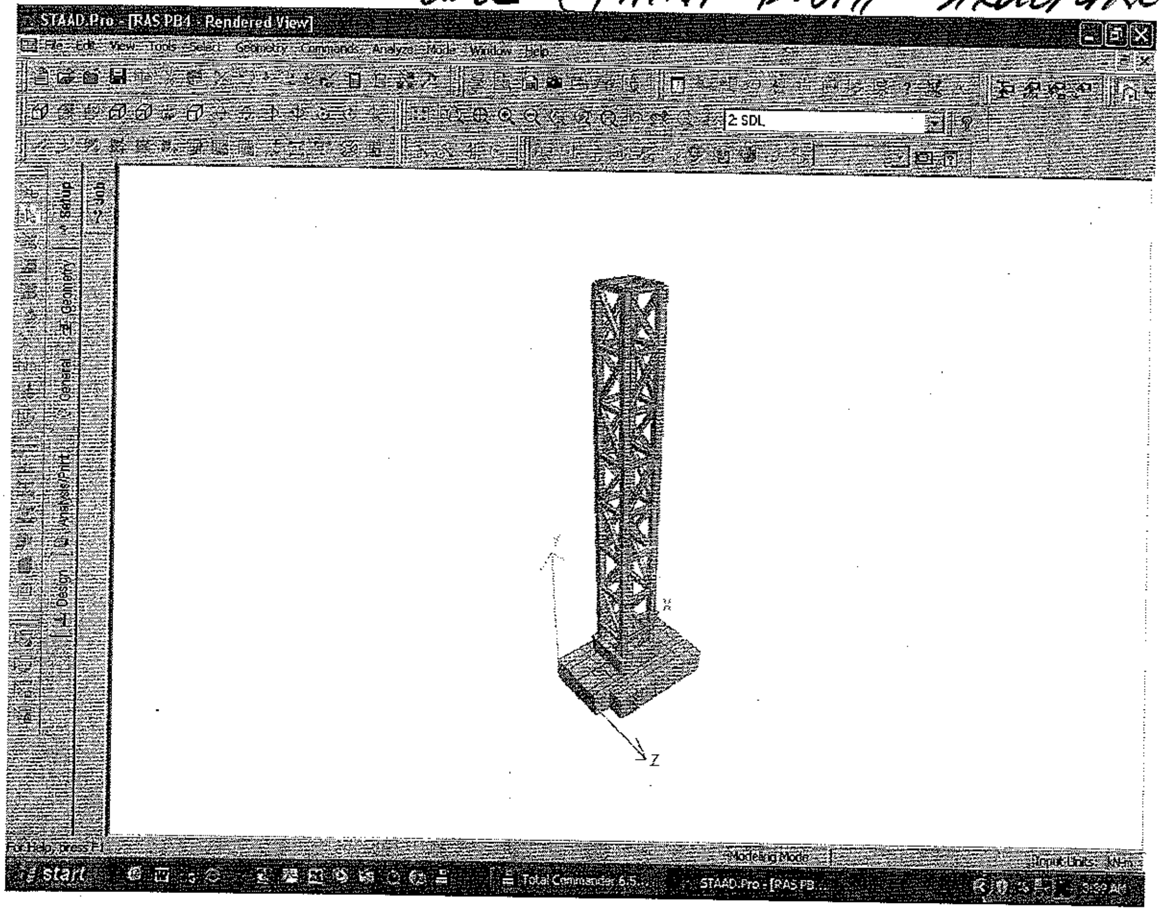
Distribution of Horozontal Forces on the Coulumn

$V_x = 0.2$ kN/m

For every 1.5m ht. interval of a column,

EQ force = 0.2 × 1.5
= 0.3 kN

STRUCTURAL MODEL (PAINT BOOTH STRUCTURE)



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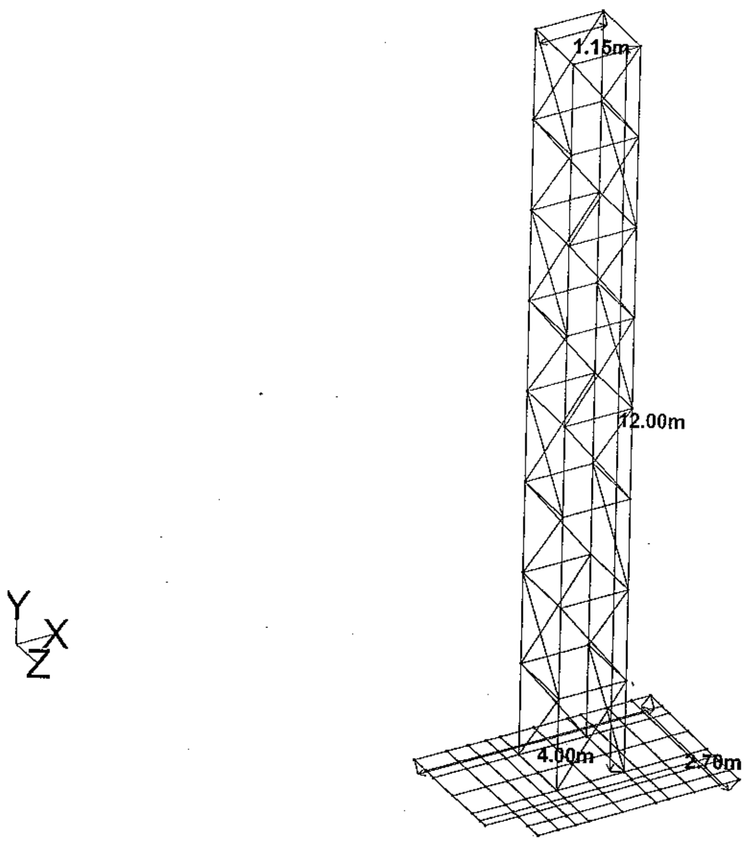
Client - GOVERNMENT OF DUBAI _ ROADS & TRANSPORT AUTHORITY

STRUCTURAL MODEL (SUPPORT FRAME and FOUNDATION)



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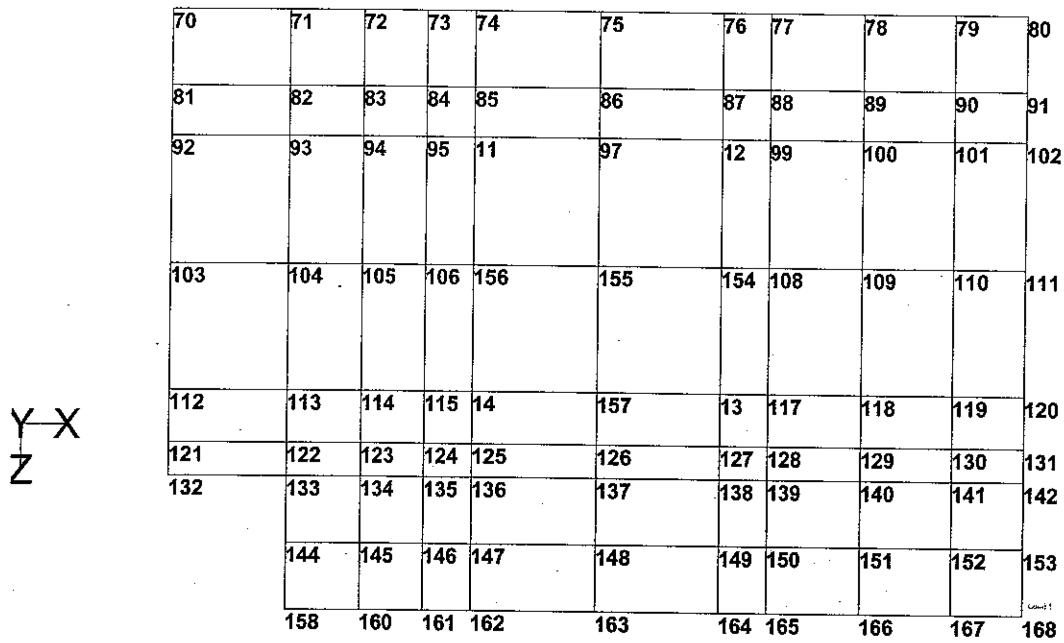

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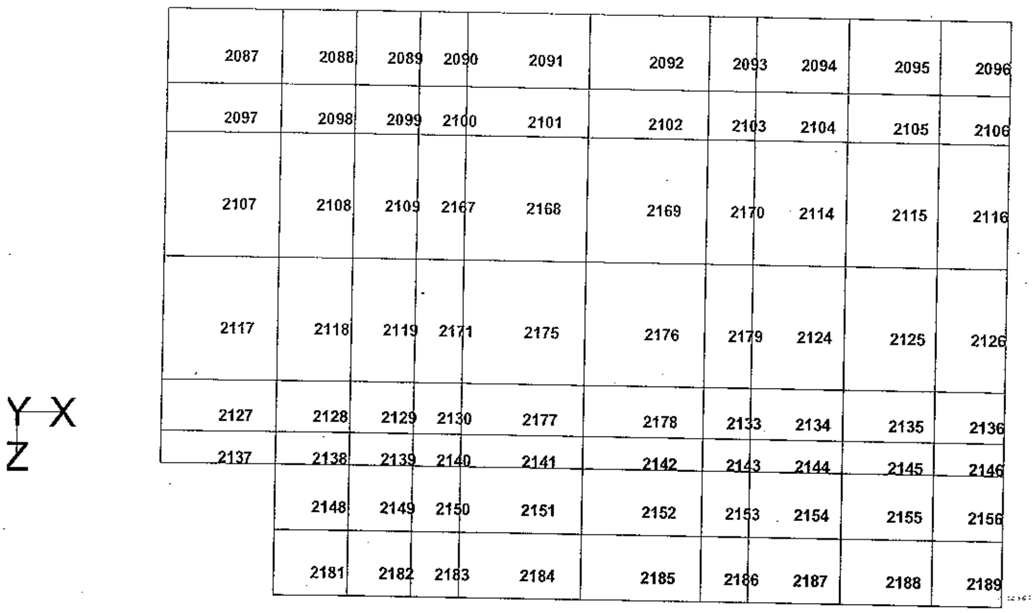

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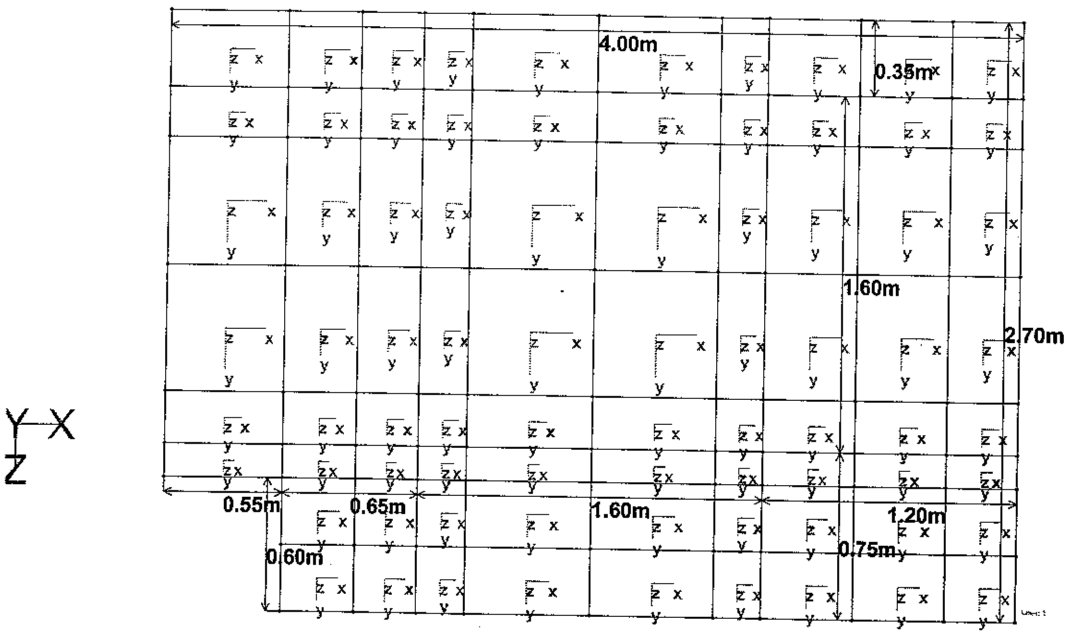
Client - GOVERNMENT OF DUBAI _ ROADS & TRANSPORT AUTHORITY

ELEMENT ORIENTATION (FOUNDATION)



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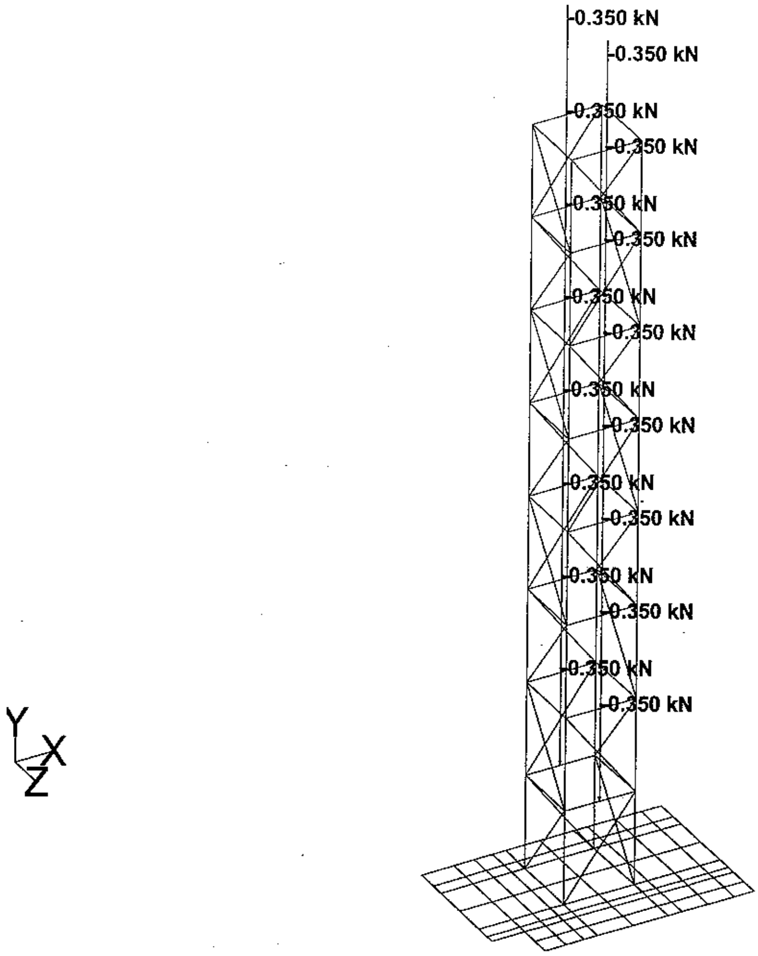

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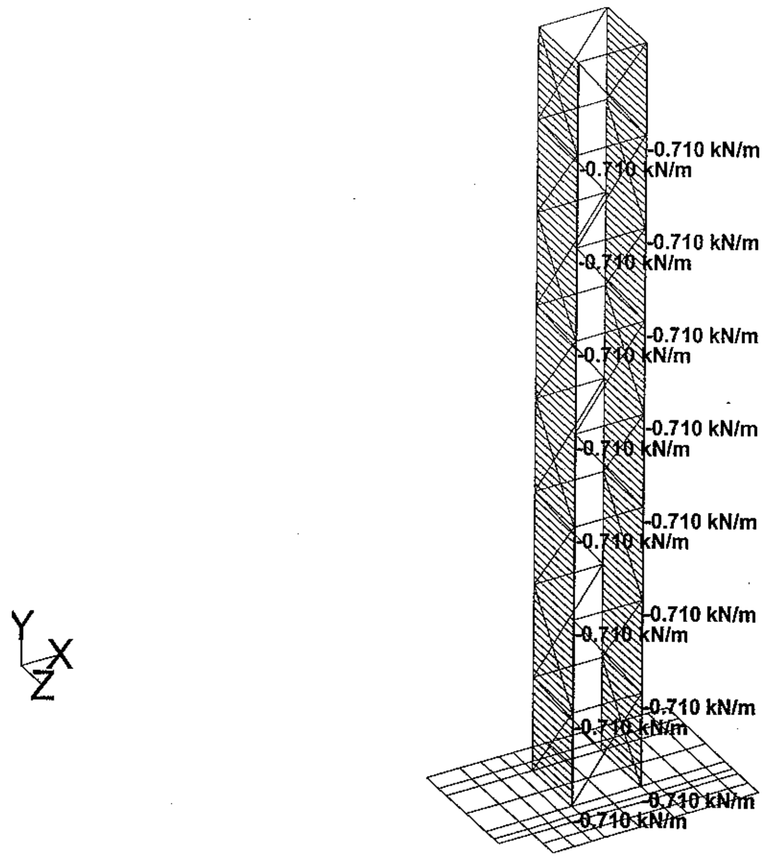

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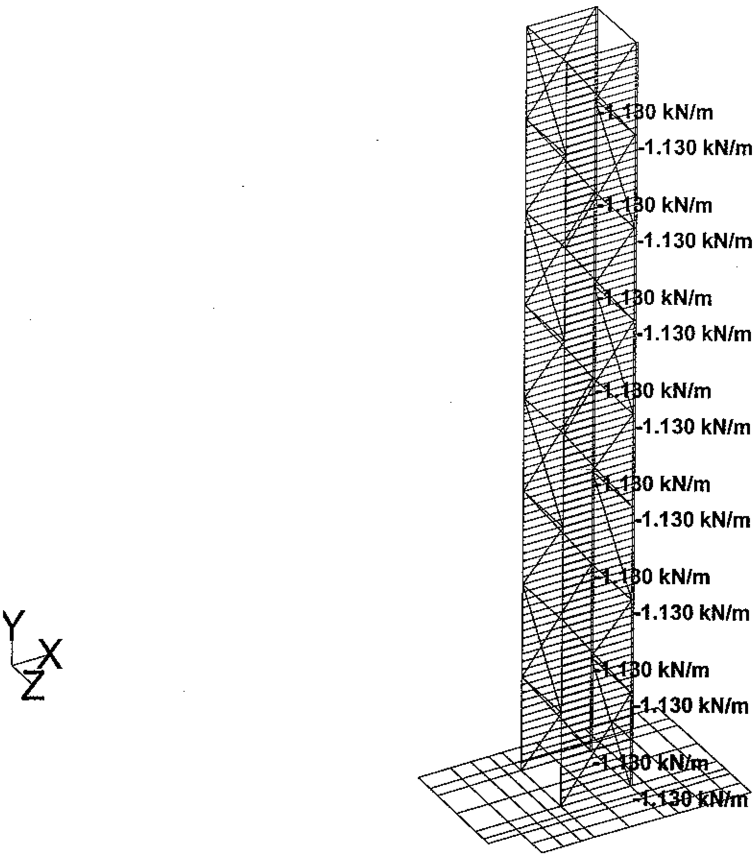

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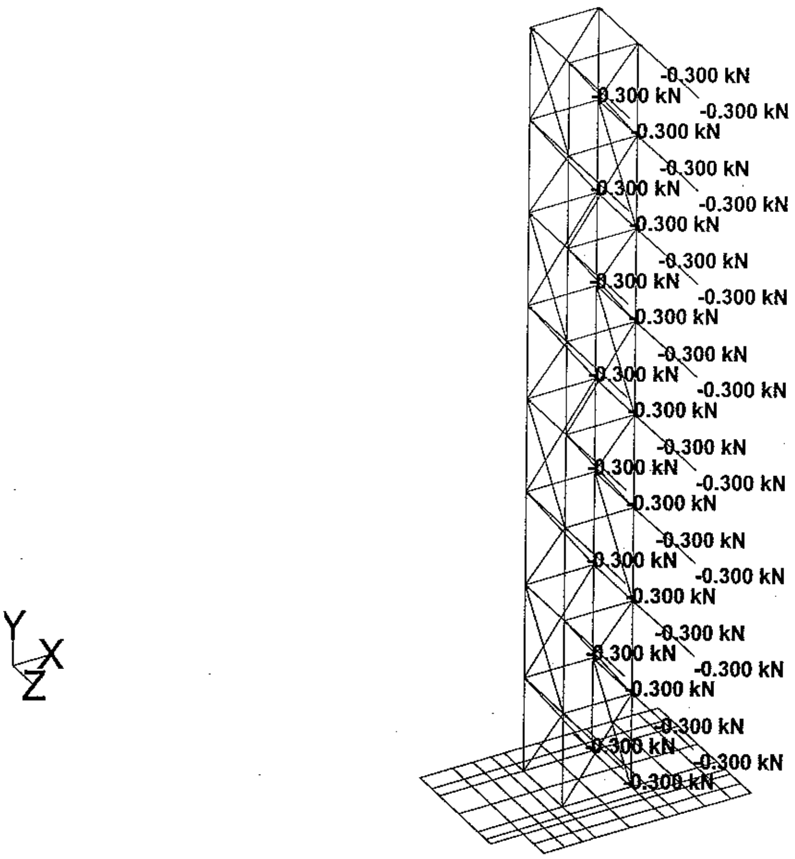

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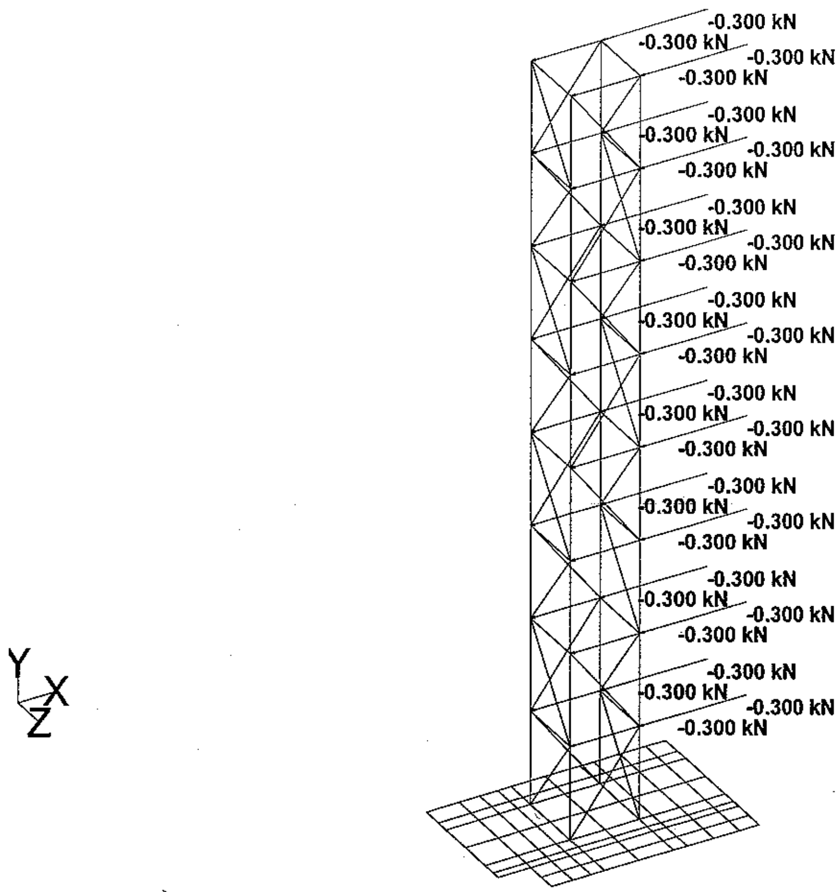

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2148 133 134 145 144; 2149 134 135 146 145; 2150 135 136 147 146
2151 136 137 148 147; 2152 137 138 149 148; 2153 138 139 150 149
2154 139 140 151 150; 2155 140 141 152 151; 2156 141 142 153 152
2167 95 11 156 106; 2168 11 97 155 156; 2169 97 12 154 155; 2170 12 99 108 154
2171 106 156 14 115; 2175 156 155 157 14; 2176 155 154 13 157
2177 14 157 126 125; 2178 157 13 127 126; 2179 154 108 117 13
2181 144 145 160 158; 2182 145 146 161 160; 2183 146 147 162 161
2184 147 148 163 162; 2185 148 149 164 163; 2186 149 150 165 164
2187 150 151 166 165; 2188 151 152 167 166; 2189 152 153 168 167
ELEMENT PROPERTY
2100 TO 2103 2130 2133 2167 TO 2171 2175 TO 2179 THICKNESS 1.95
2087 TO 2099 2104 TO 2109 2114 TO 2119 2124 TO 2129 2134 TO 2146 2148 TO 2156 -
2181 TO 2189 THICKNESS 0.75
MEMBER RELEASE
1001 TO 1004 START MX MY MZ
1101 1102 1201 1202 1301 1302 2017 TO 2022 2041 TO 2046 2065 TO 2069 -
2070 BOTH MX MY MZ
1401 1402 2023 2024 2047 2048 2071 2072 BOTH MX MY MZ
MEMBER PROPERTY BRITISH
1001 TO 1004 2001 TO 2004 2013 TO 2016 2025 TO 2028 2037 TO 2040 2049 TO 2052 -
2061 TO 2064 2073 TO 2076 TABLE ST TUB2002006.0
21 TO 24 31 TO 34 1101 1102 1201 1202 1301 1302 1401 1402 2005 TO 2012 2017 -
2018 TO 2024 2029 TO 2036 2041 TO 2048 2053 TO 2060 2065 TO 2071 -
2072 TABLE ST TUBE TH 0.0045 WT 0.125 DT 0.125
DEFINE MATERIAL START
  PAINT BOOTH DUCT SUPPORT                                    -- PAGE NO.    3

ISOTROPIC STEEL
E 2.05E+008
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2E-005
DAMP 0.03
ISOTROPIC CONCRETE
E 2.17185E+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1E-005
DAMP 0.05
END DEFINE MATERIAL
CONSTANTS
MATERIAL STEEL MEMB 21 TO 24 31 TO 34 1001 TO 1004 1101 1102 1201 1202 1301 -
1302 1401 1402 2001 TO 2076
MATERIAL CONCRETE MEMB 2087 TO 2109 2114 TO 2119 2124 TO 2130 2133 TO 2146 -
2148 TO 2156 2167 TO 2171 2175 TO 2179 2181 TO 2189
SUPPORTS
11 TO 14 70 TO 95 97 99 TO 106 108 TO 115 117 TO 142 144 TO 158 160 TO 167 -
168 ELASTIC MAT DIRECT Y SUBGRADE 18000
LOAD 1 SELFWEIGHT
SELFWEIGHT Y -1
LOAD 2 SDL
MEMBER LOAD
21 23 31 33 2005 2007 2009 2011 2029 2031 2033 2035 2053 2055 2057 -
2059 CON GY -0.35
LOAD 11 WL1
MEMBER LOAD
*WL = 1.054KPA
1001 1002 2001 2002 2013 2014 2025 2026 2037 2038 2049 2050 2061 2062 2073 -
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128. 2074 UNI GZ -0.71
129. LOAD 12 WL2
130. MEMBER LOAD
131. *WL = 1.67KPA
132. 1001 1004 2001 2004 2013 2016 2025 2028 2037 2040 2049 2052 2061 2064 2073 -
133. 2076 UNI GX -1.13
134. LOAD 21 EQ1 (EARTHQUAKE LOAD)
135. JOINT LOAD
136. 21 22 31 32 35 36 39 40 43 44 47 48 51 52 55 56 FZ -0.3
137. 23 24 33 34 37 38 41 42 45 46 49 50 53 54 57 58 FZ -0.3
138. LOAD 22 EQ2 (EARTHQUAKE LOAD)
139. JOINT LOAD
140. 21 24 31 34 35 38 39 42 43 46 47 50 51 54 55 58 FX -0.3
141. 22 23 32 33 36 37 40 41 44 45 48 49 52 53 56 57 FX -0.3
142. LOAD COMB 1001 1.4DL + 1.4SDL
143. 1 1.4 2 1.4
144. LOAD COMB 1011 1.2DL + 1.2SDL + 1.2WL1
145. 1 1.2 2 1.2 11 1.2
146. LOAD COMB 1111 1.4DL + 1.4WL1
147. 1 1.4 2 1.4 11 1.4
148. LOAD COMB 1211 1.0DL + 1.4WL1
149. 1 1.0 2 1.0 11 1.4
150. LOAD COMB 1311 1.2DL + 1.2SDL - 1.2WL1
151. 1 1.2 2 1.2 11 -1.2
152. LOAD COMB 1411 1.4DL - 1.4WL1
   PAINT BOOTH DUCT SUPPORT                                  -- PAGE NO.    4

153. 1 1.4 2 1.4 11 -1.4
154. LOAD COMB 1511 1.0DL - 1.4WL1
155. 1 1.0 2 1.0 11 -1.4
156. LOAD COMB 1012 1.2DL + 1.2SDL + 1.2WL2
157. 1 1.2 2 1.2 12 1.2
158. LOAD COMB 1112 1.4DL + 1.4WL2
159. 1 1.4 2 1.4 12 1.4
160. LOAD COMB 1212 1.0DL + 1.4WL2
161. 1 1.0 2 1.0 12 1.4
162. LOAD COMB 1312 1.2DL + 1.2SDL - 1.2WL2
163. 1 1.2 2 1.2 12 -1.2
164. LOAD COMB 1412 1.4DL - 1.4WL2
165. 1 1.4 2 1.4 12 -1.4
166. LOAD COMB 1512 1.0DL - 1.4WL2
167. 1 1.0 2 1.0 12 -1.4
168. LOAD COMB 1021 1.2DL + 1.2SDL + 1.0EQ1
169. 1 1.2 2 1.2 21 1.0
170. LOAD COMB 1121 1.2DL + 1.2SDL - 1.0EQ1
171. 1 1.2 2 1.2 21 -1.0
172. LOAD COMB 1221 0.9DL + 1.0EQ1
173. 1 0.9 2 0.9 21 1.0
174. LOAD COMB 1321 0.9DL - 1.0EQ1
175. 1 0.9 2 0.9 21 -1.0
176. LOAD COMB 1022 1.2DL + 1.2SDL + 1.0EQ2
177. 1 1.2 2 1.2 22 1.0
178. LOAD COMB 1122 1.2DL + 1.2SDL - 1.0EQ2
179. 1 1.2 2 1.2 22 -1.0
180. LOAD COMB 1222 0.9DL + 1.0EQ2
181. 1 0.9 2 0.9 22 1.0
182. LOAD COMB 1322 0.9DL - 1.0EQ2
183. 1 0.9 2 0.9 22 -1.0
184. *SLS
185. LOAD COMB 2001 1.0DL + 1.0SDL
186. 1 1.0 2 1.0
187. LOAD COMB 2011 DL + SDL + WL1
188. 1 1.0 2 1.0 11 1.0
189. LOAD COMB 2111 DL + SDL - WL1
190. 1 1.0 2 1.0 11 -1.0
191. LOAD COMB 2012 DL + SDL + WL2
192. 1 1.0 2 1.0 12 1.0
193. LOAD COMB 2112 DL + SDL - WL2
194. 1 1.0 2 1.0 12 -1.0
195. *
196. PERFORM ANALYSIS
   PAINT BOOTH DUCT SUPPORT                                  -- PAGE NO.    5
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P R O B L E M   S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 129/ 174/ 97
ORIGINAL/FINAL BAND-WIDTH= 117/ 17/ 108 DOF
TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 774
SIZE OF STIFFNESS MATRIX = 84 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 13.5/ 6713.7 MB, EXMEM = 1187.9 MB

197. LOAD LIST 2001 2011 2012 2111 2112
198. PRINT JOINT DISPLACEMENTS LIST 55
JOINT DISPLACE LIST 55
PAINT BOOTH DUCT SUPPORT -- PAGE NO. 6

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
55	2001	0.0118	-0.1754	-0.2348	-0.0002	0.0000	0.0000
	2011	0.0429	-0.2572	-1.5914	-0.0014	0.0000	0.0000
	2111	-0.0192	-0.0937	1.1219	0.0010	0.0000	0.0000
	2012	-1.2785	-0.2413	-0.2005	-0.0002	0.0000	0.0011
	2112	1.3022	-0.1096	-0.2690	-0.0002	0.0000	-0.0012

************** END OF LATEST ANALYSIS RESULT **************

199. PRINT SUPPORT REACTION ALL
SUPPORT REACTION ALL
PAINT BOOTH DUCT SUPPORT -- PAGE NO. 7

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
11	2001	0.10	4.88	0.09	0.00	0.00	0.00
	2011	0.47	6.72	8.80	0.00	0.00	0.00
	2111	-0.26	3.04	-8.62	0.00	0.00	0.00
	2012	0.46	5.92	0.66	0.00	0.00	0.00
	2112	-0.25	3.83	-0.48	0.00	0.00	0.00
12	2001	-0.10	4.91	0.09	0.00	0.00	0.00
	2011	-0.48	6.83	0.31	0.00	0.00	0.00
	2111	0.27	3.00	-0.13	0.00	0.00	0.00
	2012	13.09	3.82	-0.49	0.00	0.00	0.00
	2112	-13.29	6.01	0.67	0.00	0.00	0.00
13	2001	-0.10	4.29	-0.09	0.00	0.00	0.00
	2011	0.26	3.37	8.22	0.00	0.00	0.00
	2111	-0.47	5.21	-8.40	0.00	0.00	0.00
	2012	-0.41	3.31	0.51	0.00	0.00	0.00
	2112	0.21	5.28	-0.68	0.00	0.00	0.00
14	2001	0.10	4.26	-0.09	0.00	0.00	0.00
	2011	-0.25	3.26	-0.29	0.00	0.00	0.00
	2111	0.46	5.25	0.11	0.00	0.00	0.00
	2012	13.98	5.42	-0.67	0.00	0.00	0.00
	2112	-13.78	3.09	0.49	0.00	0.00	0.00
70	2001	0.00	1.52	0.00	0.00	0.00	0.00
	2011	0.00	2.45	0.00	0.00	0.00	0.00
	2111	0.00	0.59	0.00	0.00	0.00	0.00
	2012	0.00	2.59	0.00	0.00	0.00	0.00
	2112	0.00	0.46	0.00	0.00	0.00	0.00

(24)

71	2001	0.00	2.50	0.00	0.00	0.00	0.00
	2011	0.00	4.05	0.00	0.00	0.00	0.00
	2111	0.00	0.96	0.00	0.00	0.00	0.00
	2012	0.00	3.76	0.00	0.00	0.00	0.00
	2112	0.00	1.25	0.00	0.00	0.00	0.00
72	2001	0.00	1.81	0.00	0.00	0.00	0.00
	2011	0.00	2.94	0.00	0.00	0.00	0.00
	2111	0.00	0.69	0.00	0.00	0.00	0.00
	2012	0.00	2.50	0.00	0.00	0.00	0.00
	2112	0.00	1.13	0.00	0.00	0.00	0.00
73	2001	0.00	1.47	0.00	0.00	0.00	0.00
	2011	0.00	2.38	0.00	0.00	0.00	0.00
	2111	0.00	0.56	0.00	0.00	0.00	0.00
	2012	0.00	1.87	0.00	0.00	0.00	0.00
	2112	0.00	1.07	0.00	0.00	0.00	0.00
74	2001	0.00	2.24	0.00	0.00	0.00	0.00
	2011	0.00	3.64	0.00	0.00	0.00	0.00
	2111	0.00	0.84	0.00	0.00	0.00	0.00
	2012	0.00	2.67	0.00	0.00	0.00	0.00
	2112	0.00	1.82	0.00	0.00	0.00	0.00
75	2001	0.00	3.24	0.00	0.00	0.00	0.00
	2011	0.00	5.27	0.00	0.00	0.00	0.00
	2111	0.00	1.20	0.00	0.00	0.00	0.00

PAINT BOOTH DUCT SUPPORT -- PAGE NO. 8

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	2012	0.00	3.19	0.00	0.00	0.00	0.00
	2112	0.00	3.29	0.00	0.00	0.00	0.00
76	2001	0.00	2.26	0.00	0.00	0.00	0.00
	2011	0.00	3.69	0.00	0.00	0.00	0.00
	2111	0.00	0.83	0.00	0.00	0.00	0.00
	2012	0.00	1.76	0.00	0.00	0.00	0.00
	2112	0.00	2.75	0.00	0.00	0.00	0.00
77	2001	0.00	1.88	0.00	0.00	0.00	0.00
	2011	0.00	3.07	0.00	0.00	0.00	0.00
	2111	0.00	0.68	0.00	0.00	0.00	0.00
	2012	0.00	1.32	0.00	0.00	0.00	0.00
	2112	0.00	2.44	0.00	0.00	0.00	0.00
78	2001	0.00	2.46	0.00	0.00	0.00	0.00
	2011	0.00	4.03	0.00	0.00	0.00	0.00
	2111	0.00	0.89	0.00	0.00	0.00	0.00
	2012	0.00	1.35	0.00	0.00	0.00	0.00
	2112	0.00	3.57	0.00	0.00	0.00	0.00
79	2001	0.00	2.15	0.00	0.00	0.00	0.00
	2011	0.00	3.53	0.00	0.00	0.00	0.00
	2111	0.00	0.77	0.00	0.00	0.00	0.00
	2012	0.00	0.86	0.00	0.00	0.00	0.00
	2112	0.00	3.44	0.00	0.00	0.00	0.00
80	2001	0.00	0.94	0.00	0.00	0.00	0.00
	2011	0.00	1.54	0.00	0.00	0.00	0.00
	2111	0.00	0.33	0.00	0.00	0.00	0.00
	2012	0.00	0.27	0.00	0.00	0.00	0.00
	2112	0.00	1.60	0.00	0.00	0.00	0.00
81	2001	0.00	2.42	0.00	0.00	0.00	0.00
	2011	0.00	3.54	0.00	0.00	0.00	0.00
	2111	0.00	1.30	0.00	0.00	0.00	0.00
	2012	0.00	4.20	0.00	0.00	0.00	0.00
	2112	0.00	0.65	0.00	0.00	0.00	0.00
82	2001	0.00	3.98	0.00	0.00	0.00	0.00
	2011	0.00	5.85	0.00	0.00	0.00	0.00
	2111	0.00	2.11	0.00	0.00	0.00	0.00
	2012	0.00	6.08	0.00	0.00	0.00	0.00
	2112	0.00	1.88	0.00	0.00	0.00	0.00
83	2001	0.00	2.89	0.00	0.00	0.00	0.00
	2011	0.00	4.25	0.00	0.00	0.00	0.00
	2111	0.00	1.52	0.00	0.00	0.00	0.00
	2012	0.00	4.03	0.00	0.00	0.00	0.00

25

	2112	0.00	1.74	0.00	0.00	0.00	0.00
84	2001	0.00	2.34	0.00	0.00	0.00	0.00
	2011	0.00	3.45	0.00	0.00	0.00	0.00
	2111	0.00	1.22	0.00	0.00	0.00	0.00
	2012	0.00	3.01	0.00	0.00	0.00	0.00
	2112	0.00	1.67	0.00	0.00	0.00	0.00
85	2001	0.00	3.57	0.00	0.00	0.00	0.00

PAINT BOOTH DUCT SUPPORT -- PAGE NO. 9

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	2011	0.00	5.27	0.00	0.00	0.00	0.00
	2111	0.00	1.86	0.00	0.00	0.00	0.00
	2012	0.00	4.29	0.00	0.00	0.00	0.00
	2112	0.00	2.84	0.00	0.00	0.00	0.00
86	2001	0.00	5.15	0.00	0.00	0.00	0.00
	2011	0.00	7.64	0.00	0.00	0.00	0.00
	2111	0.00	2.66	0.00	0.00	0.00	0.00
	2012	0.00	5.10	0.00	0.00	0.00	0.00
	2112	0.00	5.19	0.00	0.00	0.00	0.00
87	2001	0.00	3.59	0.00	0.00	0.00	0.00
	2011	0.00	5.35	0.00	0.00	0.00	0.00
	2111	0.00	1.83	0.00	0.00	0.00	0.00
	2012	0.00	2.80	0.00	0.00	0.00	0.00
	2112	0.00	4.39	0.00	0.00	0.00	0.00
88	2001	0.00	2.99	0.00	0.00	0.00	0.00
	2011	0.00	4.46	0.00	0.00	0.00	0.00
	2111	0.00	1.52	0.00	0.00	0.00	0.00
	2012	0.00	2.08	0.00	0.00	0.00	0.00
	2112	0.00	3.89	0.00	0.00	0.00	0.00
89	2001	0.00	3.92	0.00	0.00	0.00	0.00
	2011	0.00	5.86	0.00	0.00	0.00	0.00
	2111	0.00	1.98	0.00	0.00	0.00	0.00
	2012	0.00	2.11	0.00	0.00	0.00	0.00
	2112	0.00	5.72	0.00	0.00	0.00	0.00
90	2001	0.00	3.42	0.00	0.00	0.00	0.00
	2011	0.00	5.13	0.00	0.00	0.00	0.00
	2111	0.00	1.72	0.00	0.00	0.00	0.00
	2012	0.00	1.32	0.00	0.00	0.00	0.00
	2112	0.00	5.53	0.00	0.00	0.00	0.00
91	2001	0.00	1.49	0.00	0.00	0.00	0.00
	2011	0.00	2.24	0.00	0.00	0.00	0.00
	2111	0.00	0.74	0.00	0.00	0.00	0.00
	2012	0.00	0.40	0.00	0.00	0.00	0.00
	2112	0.00	2.58	0.00	0.00	0.00	0.00
92	2001	0.00	3.31	0.00	0.00	0.00	0.00
	2011	0.00	4.51	0.00	0.00	0.00	0.00
	2111	0.00	2.11	0.00	0.00	0.00	0.00
	2012	0.00	5.82	0.00	0.00	0.00	0.00
	2112	0.00	0.80	0.00	0.00	0.00	0.00
93	2001	0.00	5.45	0.00	0.00	0.00	0.00
	2011	0.00	7.45	0.00	0.00	0.00	0.00
	2111	0.00	3.44	0.00	0.00	0.00	0.00
	2012	0.00	8.42	0.00	0.00	0.00	0.00
	2112	0.00	2.47	0.00	0.00	0.00	0.00
94	2001	0.00	3.95	0.00	0.00	0.00	0.00
	2011	0.00	5.41	0.00	0.00	0.00	0.00
	2111	0.00	2.48	0.00	0.00	0.00	0.00
	2012	0.00	5.58	0.00	0.00	0.00	0.00

PAINT BOOTH DUCT SUPPORT -- PAGE NO. 10

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z

26

	2112	0.00	2.32	0.00	0.00	0.00	0.00
95	2001	0.00	3.20	0.00	0.00	0.00	0.00
	2011	0.00	4.40	0.00	0.00	0.00	0.00
	2111	0.00	2.00	0.00	0.00	0.00	0.00
	2012	0.00	4.15	0.00	0.00	0.00	0.00
	2112	0.00	2.24	0.00	0.00	0.00	0.00
97	2001	0.00	7.04	0.00	0.00	0.00	0.00
	2011	0.00	9.74	0.00	0.00	0.00	0.00
	2111	0.00	4.34	0.00	0.00	0.00	0.00
	2012	0.00	7.01	0.00	0.00	0.00	0.00
	2112	0.00	7.07	0.00	0.00	0.00	0.00
99	2001	0.00	4.09	0.00	0.00	0.00	0.00
	2011	0.00	5.69	0.00	0.00	0.00	0.00
	2111	0.00	2.49	0.00	0.00	0.00	0.00
	2012	0.00	2.84	0.00	0.00	0.00	0.00
	2112	0.00	5.34	0.00	0.00	0.00	0.00
100	2001	0.00	5.36	0.00	0.00	0.00	0.00
	2011	0.00	7.48	0.00	0.00	0.00	0.00
	2111	0.00	3.24	0.00	0.00	0.00	0.00
	2012	0.00	2.85	0.00	0.00	0.00	0.00
	2112	0.00	7.87	0.00	0.00	0.00	0.00
101	2001	0.00	4.69	0.00	0.00	0.00	0.00
	2011	0.00	6.56	0.00	0.00	0.00	0.00
	2111	0.00	2.81	0.00	0.00	0.00	0.00
	2012	0.00	1.75	0.00	0.00	0.00	0.00
	2112	0.00	7.62	0.00	0.00	0.00	0.00
102	2001	0.00	2.04	0.00	0.00	0.00	0.00
	2011	0.00	2.86	0.00	0.00	0.00	0.00
	2111	0.00	1.22	0.00	0.00	0.00	0.00
	2012	0.00	0.51	0.00	0.00	0.00	0.00
	2112	0.00	3.56	0.00	0.00	0.00	0.00
103	2001	0.00	4.42	0.00	0.00	0.00	0.00
	2011	0.00	4.74	0.00	0.00	0.00	0.00
	2111	0.00	4.09	0.00	0.00	0.00	0.00
	2012	0.00	8.06	0.00	0.00	0.00	0.00
	2112	0.00	0.77	0.00	0.00	0.00	0.00
104	2001	0.00	7.27	0.00	0.00	0.00	0.00
	2011	0.00	7.86	0.00	0.00	0.00	0.00
	2111	0.00	6.68	0.00	0.00	0.00	0.00
	2012	0.00	11.62	0.00	0.00	0.00	0.00
	2112	0.00	2.92	0.00	0.00	0.00	0.00
105	2001	0.00	5.27	0.00	0.00	0.00	0.00
	2011	0.00	5.72	0.00	0.00	0.00	0.00
	2111	0.00	4.82	0.00	0.00	0.00	0.00
	2012	0.00	7.67	0.00	0.00	0.00	0.00
	2112	0.00	2.87	0.00	0.00	0.00	0.00
106	2001	0.00	4.27	0.00	0.00	0.00	0.00
	2011	0.00	4.65	0.00	0.00	0.00	0.00

PAINT BOOTH DUCT SUPPORT -- PAGE NO. 11

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	2111	0.00	3.89	0.00	0.00	0.00	0.00
	2012	0.00	5.69	0.00	0.00	0.00	0.00
	2112	0.00	2.85	0.00	0.00	0.00	0.00
108	2001	0.00	5.47	0.00	0.00	0.00	0.00
	2011	0.00	6.07	0.00	0.00	0.00	0.00
	2111	0.00	4.86	0.00	0.00	0.00	0.00
	2012	0.00	3.75	0.00	0.00	0.00	0.00
	2112	0.00	7.19	0.00	0.00	0.00	0.00
109	2001	0.00	7.16	0.00	0.00	0.00	0.00
	2011	0.00	8.00	0.00	0.00	0.00	0.00
	2111	0.00	6.33	0.00	0.00	0.00	0.00
	2012	0.00	3.66	0.00	0.00	0.00	0.00
	2112	0.00	10.67	0.00	0.00	0.00	0.00
110	2001	0.00	6.27	0.00	0.00	0.00	0.00

	2011	0.00	7.03	0.00	0.00	0.00	0.00
	2111	0.00	5.50	0.00	0.00	0.00	0.00
	2012	0.00	2.14	0.00	0.00	0.00	0.00
	2112	0.00	10.39	0.00	0.00	0.00	0.00
111	2001	0.00	2.72	0.00	0.00	0.00	0.00
	2011	0.00	3.07	0.00	0.00	0.00	0.00
	2111	0.00	2.38	0.00	0.00	0.00	0.00
	2012	0.00	0.58	0.00	0.00	0.00	0.00
	2112	0.00	4.87	0.00	0.00	0.00	0.00
112	2001	0.00	2.88	0.00	0.00	0.00	0.00
	2011	0.00	2.14	0.00	0.00	0.00	0.00
	2111	0.00	3.62	0.00	0.00	0.00	0.00
	2012	0.00	5.48	0.00	0.00	0.00	0.00
	2112	0.00	0.28	0.00	0.00	0.00	0.00
113	2001	0.00	4.75	0.00	0.00	0.00	0.00
	2011	0.00	3.57	0.00	0.00	0.00	0.00
	2111	0.00	5.92	0.00	0.00	0.00	0.00
	2012	0.00	7.87	0.00	0.00	0.00	0.00
	2112	0.00	1.63	0.00	0.00	0.00	0.00
114	2001	0.00	3.44	0.00	0.00	0.00	0.00
	2011	0.00	2.61	0.00	0.00	0.00	0.00
	2111	0.00	4.27	0.00	0.00	0.00	0.00
	2012	0.00	5.17	0.00	0.00	0.00	0.00
	2112	0.00	1.71	0.00	0.00	0.00	0.00
115	2001	0.00	2.79	0.00	0.00	0.00	0.00
	2011	0.00	2.13	0.00	0.00	0.00	0.00
	2111	0.00	3.45	0.00	0.00	0.00	0.00
	2012	0.00	3.83	0.00	0.00	0.00	0.00
	2112	0.00	1.75	0.00	0.00	0.00	0.00
117	2001	0.00	3.57	0.00	0.00	0.00	0.00
	2011	0.00	2.82	0.00	0.00	0.00	0.00
	2111	0.00	4.33	0.00	0.00	0.00	0.00
	2012	0.00	2.41	0.00	0.00	0.00	0.00
	2112	0.00	4.74	0.00	0.00	0.00	0.00

PAINT BOOTH DUCT SUPPORT -- PAGE NO. 12

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
118	2001	0.00	4.68	0.00	0.00	0.00	0.00
	2011	0.00	3.74	0.00	0.00	0.00	0.00
	2111	0.00	5.63	0.00	0.00	0.00	0.00
	2012	0.00	2.29	0.00	0.00	0.00	0.00
	2112	0.00	7.08	0.00	0.00	0.00	0.00
119	2001	0.00	4.10	0.00	0.00	0.00	0.00
	2011	0.00	3.30	0.00	0.00	0.00	0.00
	2111	0.00	4.90	0.00	0.00	0.00	0.00
	2012	0.00	1.26	0.00	0.00	0.00	0.00
	2112	0.00	6.94	0.00	0.00	0.00	0.00
120	2001	0.00	1.78	0.00	0.00	0.00	0.00
	2011	0.00	1.44	0.00	0.00	0.00	0.00
	2111	0.00	2.12	0.00	0.00	0.00	0.00
	2012	0.00	0.30	0.00	0.00	0.00	0.00
	2112	0.00	3.26	0.00	0.00	0.00	0.00
121	2001	0.00	1.33	0.00	0.00	0.00	0.00
	2011	0.00	0.79	0.00	0.00	0.00	0.00
	2111	0.00	1.88	0.00	0.00	0.00	0.00
	2012	0.00	2.58	0.00	0.00	0.00	0.00
	2112	0.00	0.08	0.00	0.00	0.00	0.00
122	2001	0.00	2.20	0.00	0.00	0.00	0.00
	2011	0.00	1.32	0.00	0.00	0.00	0.00
	2111	0.00	3.07	0.00	0.00	0.00	0.00
	2012	0.00	3.70	0.00	0.00	0.00	0.00
	2112	0.00	0.69	0.00	0.00	0.00	0.00
123	2001	0.00	1.59	0.00	0.00	0.00	0.00
	2011	0.00	0.97	0.00	0.00	0.00	0.00
	2111	0.00	2.22	0.00	0.00	0.00	0.00
	2012	0.00	2.43	0.00	0.00	0.00	0.00
	2112	0.00	0.76	0.00	0.00	0.00	0.00

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124	2001	0.00	1.29	0.00	0.00	0.00	0.00
	2011	0.00	0.79	0.00	0.00	0.00	0.00
	2111	0.00	1.79	0.00	0.00	0.00	0.00
	2012	0.00	1.79	0.00	0.00	0.00	0.00
	2112	0.00	0.79	0.00	0.00	0.00	0.00
125	2001	0.00	1.97	0.00	0.00	0.00	0.00
	2011	0.00	1.22	0.00	0.00	0.00	0.00
	2111	0.00	2.72	0.00	0.00	0.00	0.00
	2012	0.00	2.54	0.00	0.00	0.00	0.00
	2112	0.00	1.40	0.00	0.00	0.00	0.00
126	2001	0.00	2.85	0.00	0.00	0.00	0.00
	2011	0.00	1.79	0.00	0.00	0.00	0.00
	2111	0.00	3.90	0.00	0.00	0.00	0.00
	2012	0.00	2.93	0.00	0.00	0.00	0.00
	2112	0.00	2.77	0.00	0.00	0.00	0.00
127	2001	0.00	1.99	0.00	0.00	0.00	0.00
	2011	0.00	1.27	0.00	0.00	0.00	0.00
	2111	0.00	2.71	0.00	0.00	0.00	0.00

PAINT BOOTH DUCT SUPPORT -- PAGE NO. 13

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	2012	0.00	1.53	0.00	0.00	0.00	0.00
	2112	0.00	2.45	0.00	0.00	0.00	0.00
128	2001	0.00	1.66	0.00	0.00	0.00	0.00
	2011	0.00	1.07	0.00	0.00	0.00	0.00
	2111	0.00	2.25	0.00	0.00	0.00	0.00
	2012	0.00	1.11	0.00	0.00	0.00	0.00
	2112	0.00	2.20	0.00	0.00	0.00	0.00
129	2001	0.00	2.17	0.00	0.00	0.00	0.00
	2011	0.00	1.41	0.00	0.00	0.00	0.00
	2111	0.00	2.93	0.00	0.00	0.00	0.00
	2012	0.00	1.04	0.00	0.00	0.00	0.00
	2112	0.00	3.30	0.00	0.00	0.00	0.00
130	2001	0.00	1.90	0.00	0.00	0.00	0.00
	2011	0.00	1.25	0.00	0.00	0.00	0.00
	2111	0.00	2.54	0.00	0.00	0.00	0.00
	2012	0.00	0.55	0.00	0.00	0.00	0.00
	2112	0.00	3.25	0.00	0.00	0.00	0.00
131	2001	0.00	0.83	0.00	0.00	0.00	0.00
	2011	0.00	0.55	0.00	0.00	0.00	0.00
	2111	0.00	1.10	0.00	0.00	0.00	0.00
	2012	0.00	0.12	0.00	0.00	0.00	0.00
	2112	0.00	1.53	0.00	0.00	0.00	0.00
132	2001	0.00	1.43	0.00	0.00	0.00	0.00
	2011	0.00	0.70	0.00	0.00	0.00	0.00
	2111	0.00	2.16	0.00	0.00	0.00	0.00
	2012	0.00	2.81	0.00	0.00	0.00	0.00
	2112	0.00	0.06	0.00	0.00	0.00	0.00
133	2001	0.00	2.51	0.00	0.00	0.00	0.00
	2011	0.00	1.26	0.00	0.00	0.00	0.00
	2111	0.00	3.77	0.00	0.00	0.00	0.00
	2012	0.00	4.28	0.00	0.00	0.00	0.00
	2112	0.00	0.75	0.00	0.00	0.00	0.00
134	2001	0.00	1.82	0.00	0.00	0.00	0.00
	2011	0.00	0.93	0.00	0.00	0.00	0.00
	2111	0.00	2.72	0.00	0.00	0.00	0.00
	2012	0.00	2.81	0.00	0.00	0.00	0.00
	2112	0.00	0.84	0.00	0.00	0.00	0.00
135	2001	0.00	1.48	0.00	0.00	0.00	0.00
	2011	0.00	0.76	0.00	0.00	0.00	0.00
	2111	0.00	2.20	0.00	0.00	0.00	0.00
	2012	0.00	2.07	0.00	0.00	0.00	0.00
	2112	0.00	0.88	0.00	0.00	0.00	0.00
136	2001	0.00	2.26	0.00	0.00	0.00	0.00
	2011	0.00	1.17	0.00	0.00	0.00	0.00
	2111	0.00	3.34	0.00	0.00	0.00	0.00
	2012	0.00	2.93	0.00	0.00	0.00	0.00

	2112	0.00	1.59	0.00	0.00	0.00	0.00
137	2001	0.00	3.26	0.00	0.00	0.00	0.00

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	2011	0.00	1.73	0.00	0.00	0.00	0.00
	2111	0.00	4.79	0.00	0.00	0.00	0.00
	2012	0.00	3.36	0.00	0.00	0.00	0.00
	2112	0.00	3.15	0.00	0.00	0.00	0.00
138	2001	0.00	2.28	0.00	0.00	0.00	0.00
	2011	0.00	1.23	0.00	0.00	0.00	0.00
	2111	0.00	3.32	0.00	0.00	0.00	0.00
	2012	0.00	1.75	0.00	0.00	0.00	0.00
	2112	0.00	2.80	0.00	0.00	0.00	0.00
139	2001	0.00	1.90	0.00	0.00	0.00	0.00
	2011	0.00	1.03	0.00	0.00	0.00	0.00
	2111	0.00	2.76	0.00	0.00	0.00	0.00
	2012	0.00	1.27	0.00	0.00	0.00	0.00
	2112	0.00	2.52	0.00	0.00	0.00	0.00
140	2001	0.00	2.48	0.00	0.00	0.00	0.00
	2011	0.00	1.38	0.00	0.00	0.00	0.00
	2111	0.00	3.59	0.00	0.00	0.00	0.00
	2012	0.00	1.17	0.00	0.00	0.00	0.00
	2112	0.00	3.80	0.00	0.00	0.00	0.00
141	2001	0.00	2.17	0.00	0.00	0.00	0.00
	2011	0.00	1.22	0.00	0.00	0.00	0.00
	2111	0.00	3.13	0.00	0.00	0.00	0.00
	2012	0.00	0.61	0.00	0.00	0.00	0.00
	2112	0.00	3.74	0.00	0.00	0.00	0.00
142	2001	0.00	0.95	0.00	0.00	0.00	0.00
	2011	0.00	0.54	0.00	0.00	0.00	0.00
	2111	0.00	1.35	0.00	0.00	0.00	0.00
	2012	0.00	0.13	0.00	0.00	0.00	0.00
	2112	0.00	1.76	0.00	0.00	0.00	0.00
144	2001	0.00	3.06	0.00	0.00	0.00	0.00
	2011	0.00	0.87	0.00	0.00	0.00	0.00
	2111	0.00	5.24	0.00	0.00	0.00	0.00
	2012	0.00	5.32	0.00	0.00	0.00	0.00
	2112	0.00	0.79	0.00	0.00	0.00	0.00
145	2001	0.00	2.33	0.00	0.00	0.00	0.00
	2011	0.00	0.68	0.00	0.00	0.00	0.00
	2111	0.00	3.97	0.00	0.00	0.00	0.00
	2012	0.00	3.66	0.00	0.00	0.00	0.00
	2112	0.00	1.00	0.00	0.00	0.00	0.00
146	2001	0.00	1.89	0.00	0.00	0.00	0.00
	2011	0.00	0.57	0.00	0.00	0.00	0.00
	2111	0.00	3.21	0.00	0.00	0.00	0.00
	2012	0.00	2.69	0.00	0.00	0.00	0.00
	2112	0.00	1.08	0.00	0.00	0.00	0.00
147	2001	0.00	2.88	0.00	0.00	0.00	0.00
	2011	0.00	0.88	0.00	0.00	0.00	0.00
	2111	0.00	4.88	0.00	0.00	0.00	0.00
	2012	0.00	3.80	0.00	0.00	0.00	0.00

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	2112	0.00	1.96	0.00	0.00	0.00	0.00
148	2001	0.00	4.16	0.00	0.00	0.00	0.00
	2011	0.00	1.32	0.00	0.00	0.00	0.00

	2111	0.00	7.00	0.00	0.00	0.00	0.00
	2012	0.00	4.33	0.00	0.00	0.00	0.00
	2112	0.00	3.99	0.00	0.00	0.00	0.00
149	2001	0.00	2.91	0.00	0.00	0.00	0.00
	2011	0.00	0.96	0.00	0.00	0.00	0.00
	2111	0.00	4.85	0.00	0.00	0.00	0.00
	2012	0.00	2.23	0.00	0.00	0.00	0.00
	2112	0.00	3.58	0.00	0.00	0.00	0.00
150	2001	0.00	2.42	0.00	0.00	0.00	0.00
	2011	0.00	0.81	0.00	0.00	0.00	0.00
	2111	0.00	4.03	0.00	0.00	0.00	0.00
	2012	0.00	1.60	0.00	0.00	0.00	0.00
	2112	0.00	3.24	0.00	0.00	0.00	0.00
151	2001	0.00	3.17	0.00	0.00	0.00	0.00
	2011	0.00	1.10	0.00	0.00	0.00	0.00
	2111	0.00	5.25	0.00	0.00	0.00	0.00
	2012	0.00	1.45	0.00	0.00	0.00	0.00
	2112	0.00	4.90	0.00	0.00	0.00	0.00
152	2001	0.00	2.78	0.00	0.00	0.00	0.00
	2011	0.00	0.99	0.00	0.00	0.00	0.00
	2111	0.00	4.57	0.00	0.00	0.00	0.00
	2012	0.00	0.71	0.00	0.00	0.00	0.00
	2112	0.00	4.84	0.00	0.00	0.00	0.00
153	2001	0.00	1.21	0.00	0.00	0.00	0.00
	2011	0.00	0.44	0.00	0.00	0.00	0.00
	2111	0.00	1.98	0.00	0.00	0.00	0.00
	2012	0.00	0.13	0.00	0.00	0.00	0.00
	2112	0.00	2.29	0.00	0.00	0.00	0.00
154	2001	0.00	6.57	0.00	0.00	0.00	0.00
	2011	0.00	7.27	0.00	0.00	0.00	0.00
	2111	0.00	5.86	0.00	0.00	0.00	0.00
	2012	0.00	5.09	0.00	0.00	0.00	0.00
	2112	0.00	8.04	0.00	0.00	0.00	0.00
155	2001	0.00	9.41	0.00	0.00	0.00	0.00
	2011	0.00	10.35	0.00	0.00	0.00	0.00
	2111	0.00	8.47	0.00	0.00	0.00	0.00
	2012	0.00	9.48	0.00	0.00	0.00	0.00
	2112	0.00	9.34	0.00	0.00	0.00	0.00
156	2001	0.00	6.52	0.00	0.00	0.00	0.00
	2011	0.00	7.12	0.00	0.00	0.00	0.00
	2111	0.00	5.91	0.00	0.00	0.00	0.00
	2012	0.00	8.09	0.00	0.00	0.00	0.00
	2112	0.00	4.95	0.00	0.00	0.00	0.00
157	2001	0.00	6.15	0.00	0.00	0.00	0.00
	2011	0.00	4.77	0.00	0.00	0.00	0.00

PAINT BOOTH DUCT SUPPORT -- PAGE NO. 16

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	2111	0.00	7.52	0.00	0.00	0.00	0.00
	2012	0.00	6.28	0.00	0.00	0.00	0.00
	2112	0.00	6.01	0.00	0.00	0.00	0.00
158	2001	0.00	0.83	0.00	0.00	0.00	0.00
	2011	0.00	0.04	0.00	0.00	0.00	0.00
	2111	0.00	1.62	0.00	0.00	0.00	0.00
	2012	0.00	1.48	0.00	0.00	0.00	0.00
	2112	0.00	0.18	0.00	0.00	0.00	0.00
160	2001	0.00	1.11	0.00	0.00	0.00	0.00
	2011	0.00	0.07	0.00	0.00	0.00	0.00
	2111	0.00	2.16	0.00	0.00	0.00	0.00
	2012	0.00	1.79	0.00	0.00	0.00	0.00
	2112	0.00	0.44	0.00	0.00	0.00	0.00
161	2001	0.00	0.90	0.00	0.00	0.00	0.00
	2011	0.00	0.06	0.00	0.00	0.00	0.00
	2111	0.00	1.74	0.00	0.00	0.00	0.00
	2012	0.00	1.31	0.00	0.00	0.00	0.00
	2112	0.00	0.49	0.00	0.00	0.00	0.00
162	2001	0.00	1.38	0.00	0.00	0.00	0.00

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	2011	0.00	0.10	0.00	0.00	0.00	0.00
	2111	0.00	2.65	0.00	0.00	0.00	0.00
	2012	0.00	1.84	0.00	0.00	0.00	0.00
	2112	0.00	0.91	0.00	0.00	0.00	0.00
163	2001	0.00	1.99	0.00	0.00	0.00	0.00
	2011	0.00	0.17	0.00	0.00	0.00	0.00
	2111	0.00	3.81	0.00	0.00	0.00	0.00
	2012	0.00	2.09	0.00	0.00	0.00	0.00
	2112	0.00	1.89	0.00	0.00	0.00	0.00
164	2001	0.00	1.39	0.00	0.00	0.00	0.00
	2011	0.00	0.14	0.00	0.00	0.00	0.00
	2111	0.00	2.64	0.00	0.00	0.00	0.00
	2012	0.00	1.06	0.00	0.00	0.00	0.00
	2112	0.00	1.72	0.00	0.00	0.00	0.00
165	2001	0.00	1.16	0.00	0.00	0.00	0.00
	2011	0.00	0.12	0.00	0.00	0.00	0.00
	2111	0.00	2.19	0.00	0.00	0.00	0.00
	2012	0.00	0.76	0.00	0.00	0.00	0.00
	2112	0.00	1.56	0.00	0.00	0.00	0.00
166	2001	0.00	1.52	0.00	0.00	0.00	0.00
	2011	0.00	0.18	0.00	0.00	0.00	0.00
	2111	0.00	2.86	0.00	0.00	0.00	0.00
	2012	0.00	0.67	0.00	0.00	0.00	0.00
	2112	0.00	2.37	0.00	0.00	0.00	0.00
167	2001	0.00	1.33	0.00	0.00	0.00	0.00
	2011	0.00	0.17	0.00	0.00	0.00	0.00
	2111	0.00	2.49	0.00	0.00	0.00	0.00
	2012	0.00	0.31	0.00	0.00	0.00	0.00
	2112	0.00	2.35	0.00	0.00	0.00	0.00

PAINT BOOTH DUCT SUPPORT -- PAGE NO. 17

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
168	2001	0.00	0.58	0.00	0.00	0.00	0.00
	2011	0.00	0.08	0.00	0.00	0.00	0.00
	2111	0.00	1.08	0.00	0.00	0.00	0.00
	2012	0.00	0.04	0.00	0.00	0.00	0.00
	2112	0.00	1.11	0.00	0.00	0.00	0.00

************** END OF LATEST ANALYSIS RESULT **************

```
200. LOAD LIST 1001 1011 1012 1021 1022 1111 1112 1121 1122 1211 1212 1221 1222 -
201. 1311 1312 1321 1322 1411 1412 1511 1512
202. PARAMETER
203. CODE BS5950
204. PY 345000 MEMB 21 TO 24 31 TO 34 1001 TO 1004 1101 1102 1201 1202 1301 1302 -
205. 1401 1402 2001 TO 2076
206. *TRACK 2 ALL
207. CHECK CODE ALL
```

STEEL DESIGN

STAAD.Pro CODE CHECKING - (BSI)
****

PROGRAM CODE REVISION V2.9_5950-1_2000

PAINT BOOTH DUCT SUPPORT -- PAGE NO. 18

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION

(32)

21	ST TUB E	PASS	BS-4.8.2.2	0.020	1412
		1.41 T	0.03	0.59	1.15
22	ST TUB E	PASS	BS-4.8.2.2	0.012	1411
		1.03 T	0.03	0.34	1.14
23	ST TUB E	PASS	BS-4.8.2.2	0.023	1112
		3.96 T	0.04	0.57	1.15
24	ST TUB E	PASS	BS-4.8.2.2	0.014	1111
		2.62 T	0.03	0.33	1.14
31	ST TUB E	PASS	BS-4.8.2.2	0.022	1112
		2.45 T	-0.05	-0.59	0.00
32	ST TUB E	PASS	BS-4.8.2.2	0.014	1111
		1.74 T	-0.05	-0.34	0.00
33	ST TUB E	PASS	BS-4.8.2.2	0.020	1112
		0.73 T	-0.03	0.60	1.15
34	ST TUB E	PASS	BS-4.8.2.2	0.012	1111
		0.66 T	-0.03	0.35	1.14
1001	ST TUB 2002006.0	PASS	BS-4.8.3.2	0.088	1112
		109.11 C	-1.37	-0.82	1.50
1002	ST TUB 2002006.0	PASS	BS-4.8.3.2	0.074	1412
		86.59 C	1.41	0.75	1.50
1003	ST TUB 2002006.0	PASS	BS-4.8.3.2	0.085	1412
		109.36 C	1.37	0.44	1.50
1004	ST TUB 2002006.0	PASS	BS-4.8.3.2	0.077	1112
		86.90 C	-1.41	-1.08	1.50
1101	ST TUB E	PASS	BS-4.7 (C)	0.028	1111
		20.08 C	0.00	0.00	1.89
1102	ST TUB E	PASS	BS-4.7 (C)	0.022	1411
		15.86 C	0.00	0.00	0.00
1201	ST TUB E	PASS	BS-4.7 (C)	0.029	1411
		20.16 C	0.00	0.00	1.89
1202	ST TUB E	PASS	BS-4.7 (C)	0.022	1111
		15.71 C	0.00	0.00	0.00
1301	ST TUB E	PASS	BS-4.7 (C)	0.045	1112
		31.59 C	0.00	0.00	1.89
1302	ST TUB E	PASS	BS-4.7 (C)	0.035	1412
		24.56 C	0.00	0.00	0.00
1401	ST TUB E	PASS	BS-4.7 (C)	0.045	1412
		31.59 C	0.00	0.00	1.89
1402	ST TUB E	PASS	BS-4.7 (C)	0.034	1112
		24.26 C	0.00	0.00	0.00
2001	ST TUB 2002006.0	PASS	BS-4.8.3.2	0.065	1112
		63.99 C	1.33	1.40	0.00
2002	ST TUB 2002006.0	PASS	BS-4.8.3.2	0.076	1412
		82.57 C	-1.32	-1.34	0.00
2003	ST TUB 2002006.0	PASS	BS-4.8.3.2	0.061	1412
		64.02 C	-1.33	-1.02	0.00
2004	ST TUB 2002006.0	PASS	BS-4.8.3.2	0.079	1112
		82.78 C	1.32	1.65	0.00

PAINT BOOTH DUCT SUPPORT -- PAGE NO. 19

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
2005	ST TUB E	PASS	BS-4.8.2.2	0.016	1112
		0.54 T	0.02	-0.51	0.00
2006	ST TUB E	PASS	BS-4.8.2.2	0.010	1111
		0.50 T	0.03	-0.29	0.00
2007	ST TUB E	PASS	BS-4.8.2.2	0.019	1112
		2.56 T	0.04	0.50	1.15
2008	ST TUB E	PASS	BS-4.8.2.2	0.012	1111
		1.77 T	0.03	0.28	1.14
2009	ST TUB E	PASS	BS-4.8.2.2	0.016	1112
		2.46 T	-0.03	-0.41	0.00
2010	ST TUB E	PASS	BS-4.8.2.2	0.010	1111
		1.68 T	-0.03	-0.23	0.00
2011	ST TUB E	PASS	BS-4.8.2.2	0.014	1112

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                         0.46 T         -0.02           0.43         1.15
    2012 ST TUB E         PASS       BS-4.8.2.2        0.008        1111
                         0.42 T         -0.02           0.24         1.14
    2013 ST TUB 2002006.0 PASS       BS-4.8.3.2        0.059        1112
                        62.23 C         -1.10           1.16         0.00
    2014 ST TUB 2002006.0 PASS       BS-4.8.3.2        0.044        1412
                        45.88 C          1.11          -0.62         0.00
    2015 ST TUB 2002006.0 PASS       BS-4.8.3.2        0.057        1412
                        62.19 C          1.10          -0.92         0.00
    2016 ST TUB 2002006.0 PASS       BS-4.8.3.2        0.048        1112
                        46.03 C         -1.11           0.96         0.00
    2017 ST TUB E         PASS       BS-4.7    (C)     0.019        1111
                        13.53 C          0.00           0.00         1.89
    2018 ST TUB E         PASS       BS-4.7    (C)     0.016        1411
                        11.25 C          0.00           0.00         0.00
    2019 ST TUB E         PASS       BS-4.7    (C)     0.019        1411
                        13.70 C          0.00           0.00         1.89
    2020 ST TUB E         PASS       BS-4.7    (C)     0.016        1111
                        11.09 C          0.00           0.00         0.00
    2021 ST TUB E         PASS       BS-4.7    (C)     0.030        1112
                        20.95 C          0.00           0.00         1.89
    2022 ST TUB E         PASS       BS-4.7    (C)     0.025        1412
                        17.38 C          0.00           0.00         0.00
    2023 ST TUB E         PASS       BS-4.7    (C)     0.030        1412
                        21.17 C          0.00           0.00         1.89
    2024 ST TUB E         PASS       BS-4.7    (C)     0.024        1112
                        17.11 C          0.00           0.00         0.00
    2025 ST TUB 2002006.0 PASS       BS-4.8.3.2        0.034        1112
                        31.14 C          0.81           0.78         0.00
    2026 ST TUB 2002006.0 PASS       BS-4.8.3.2        0.041        1412
                        44.34 C         -0.79          -0.70         0.00
    2027 ST TUB 2002006.0 PASS       BS-4.8.3.2        0.031        1412
                        31.05 C         -0.81          -0.43         0.00
      PAINT BOOTH DUCT SUPPORT                                   -- PAGE NO.    20


ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE          RESULT/   CRITICAL COND/      RATIO/     LOADING/
                            FX            MY              MZ       LOCATION
=======================================================================


    2028 ST TUB 2002006.0 PASS       BS-4.8.3.2        0.044        1112
                        44.45 C          0.79           0.94         0.00
    2029 ST TUB E         PASS       BS-4.8.2.2        0.011        1112
                         0.38 T          0.02          -0.34         0.00
    2030 ST TUB E         PASS       BS-4.8.2.2        0.007        1111
                         0.34 T          0.02          -0.19         0.00
    2031 ST TUB E         PASS       BS-4.8.2.2        0.014        1112
                         2.38 T          0.02           0.33         1.15
    2032 ST TUB E         PASS       BS-4.8.2.2        0.008        1111
                         1.60 T          0.02           0.17         1.14
    2033 ST TUB E         PASS       BS-4.8.2.2        0.011        1112
                         2.27 T         -0.02          -0.25         0.00
    2034 ST TUB E         PASS       BS-4.8.2.2        0.006        1111
                         1.50 T         -0.01          -0.12         0.00
    2035 ST TUB E         PASS       BS-4.8.2.2        0.008        1112
                         0.26 T         -0.01           0.26         1.15
    2036 ST TUB E         PASS       BS-4.8.2.2        0.005        1111
                         0.23 T         -0.01           0.13         1.14
    2037 ST TUB 2002006.0 PASS       BS-4.8.3.2        0.030        1112
                        29.61 C         -0.53           0.73         0.00
    2038 ST TUB 2002006.0 PASS       BS-4.8.3.2        0.021        1111
                        20.96 C          0.49           0.33         0.00
    2039 ST TUB 2002006.0 PASS       BS-4.8.3.2        0.028        1412
                        29.51 C          0.53          -0.48         0.00
    2040 ST TUB 2002006.0 PASS       BS-4.8.3.2        0.023        1112
                        19.26 C         -0.55           0.62         0.00
    2041 ST TUB E         PASS       BS-4.7    (C)     0.012        1111
                         8.65 C          0.00           0.00         1.89
    2042 ST TUB E         PASS       BS-4.8.2.2        0.009        1111
                         5.45 T          0.00          -0.06         0.94
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2043	ST TUB E	PASS	BS-4.7 (C)	0.012	1411
		8.81 C	0.00	0.00	1.89
2044	ST TUB E	PASS	BS-4.8.2.2	0.009	1411
		5.30 T	0.00	-0.06	0.94
2045	ST TUB E	PASS	BS-4.7 (C)	0.019	1112
		13.33 C	0.00	0.00	1.89
2046	ST TUB E	PASS	BS-4.7 (C)	0.014	1412
		9.80 C	0.00	0.00	0.00
2047	ST TUB E	PASS	BS-4.7 (C)	0.019	1412
		13.59 C	0.00	0.00	1.89
2048	ST TUB E	PASS	BS-4.7 (C)	0.014	1112
		9.54 C	0.00	0.00	0.00
2049	ST TUB 2002006.0	PASS	BS-4.8.3.2	0.014	1112
		10.32 C	0.34	0.50	0.00
2050	ST TUB 2002006.0	PASS	BS-4.8.3.2	0.017	1412
		17.61 C	-0.32	-0.30	0.00

PAINT BOOTH DUCT SUPPORT -- PAGE NO. 21

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
2051	ST TUB 2002006.0	PASS	BS-4.8.3.2	0.012	1411
		12.80 C	-0.21	0.19	0.00
2052	ST TUB 2002006.0	PASS	BS-4.8.3.2	0.019	1112
		17.70 C	0.32	0.55	0.00
2053	ST TUB E	PASS	BS-4.8.2.2	0.006	1112
		0.32 T	-0.01	0.18	0.00
2054	ST TUB E	PASS	BS-4.8.2.2	0.003	1111
		0.22 T	-0.01	0.08	0.00
2055	ST TUB E	PASS	BS-4.8.2.2	0.009	1112
		2.45 T	-0.01	-0.18	1.15
2056	ST TUB E	PASS	BS-4.8.2.2	0.005	1111
		1.56 T	-0.01	-0.08	1.14
2057	ST TUB E	PASS	BS-4.8.3.3.3	0.005	1412
		1.03 C	0.00	0.11	-
2058	ST TUB E	PASS	BS-4.8.3.3.3	0.002	1411
		0.60 C	0.00	0.04	-
2059	ST TUB E	PASS	BS-4.8.3.3.3	0.004	1412
		0.03 C	0.00	0.14	-
2060	ST TUB E	PASS	BS-4.8.2.2	0.002	1411
		0.02 T	0.00	-0.06	1.14
2061	ST TUB 2002006.0	PASS	BS-4.8.3.2	0.011	1112
		8.75 C	-0.16	0.39	0.00
2062	ST TUB 2002006.0	PASS	BS-4.8.3.2	0.007	1111
		6.63 C	0.25	0.09	0.00
2063	ST TUB 2002006.0	PASS	BS-4.8.3.2	0.008	1412
		8.66 C	0.16	-0.16	0.00
2064	ST TUB 2002006.0	PASS	BS-4.8.3.2	0.008	1112
		4.32 C	-0.19	0.40	0.00
2065	ST TUB E	PASS	BS-4.8.2.2	0.006	1411
		3.14 T	0.00	-0.06	0.94
2066	ST TUB E	PASS	BS-4.8.2.2	0.003	1111
		1.05 T	0.00	-0.06	0.94
2067	ST TUB E	PASS	BS-4.8.2.2	0.006	1111
		3.30 T	0.00	-0.06	0.94
2068	ST TUB E	PASS	BS-4.8.2.2	0.003	1411
		0.95 T	0.00	-0.06	0.94
2069	ST TUB E	PASS	BS-4.8.2.2	0.009	1412
		5.12 T	0.00	-0.06	0.95
2070	ST TUB E	PASS	BS-4.8.2.2	0.004	1112
		1.72 T	0.00	-0.06	0.95
2071	ST TUB E	PASS	BS-4.8.2.2	0.009	1112
		5.38 T	0.00	-0.06	0.95
2072	ST TUB E	PASS	BS-4.8.2.2	0.004	1412
		1.56 T	0.00	-0.06	0.95
2073	ST TUB 2002006.0	PASS	BS-4.8.3.2	0.005	1112
		1.30 C	0.09	0.37	0.00

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ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
2074	ST TUB 2002006.0	PASS	BS-4.8.3.2	0.004	1111
		1.34 C	0.20	-0.11	0.00
2075	ST TUB 2002006.0	PASS	BS-4.8.3.3.3	0.002	1411
		2.21 C	0.03	0.07	-
2076	ST TUB 2002006.0	PASS	BS-4.8.3.2	0.005	1112
		2.77 C	0.06	0.34	0.00

************** END OF TABULATED RESULT OF DESIGN **************

208. START CONCRETE DESIGN
PAINT BOOTH DUCT SUPPORT -- PAGE NO. 23

CONCRETE DESIGN
209. CODE BS8110
PROGRAM CODE REVISION V1.5_8110_97/1
210. FYMAIN 460000 MEMB 2087 TO 2109 2114 TO 2119 2124 TO 2130 2133 TO 2146 2148 -
211. 2149 TO 2156 2167 TO 2171 2175 TO 2179
212. FC 40000 MEMB 2087 TO 2109 2114 TO 2119 2124 TO 2130 2133 TO 2146 -
213. 2148 TO 2156 2167 TO 2171 2175 TO 2179
214. CLEAR 0.075 MEMB 2087 TO 2109 2114 TO 2119 2124 TO 2130 2133 TO 2146 2148 -
215. 2149 TO 2156 2167 TO 2171 2175 TO 2179
216. DESIGN ELEMENT 2087 TO 2099 2104 TO 2109 2114 TO 2119 2124 TO 2129 -
217. 2134 TO 2146 2148 TO 2156 2181 TO 2189
PAINT BOOTH DUCT SUPPORT -- PAGE NO. 24

ELEMENT DESIGN SUMMARY-BASED ON 16mm BARS

MINIMUM AREAS ARE ACTUAL CODE MIN % REQUIREMENTS.
PRACTICAL LAYOUTS ARE AS FOLLOWS:
FY=460, 6No.16mm BARS AT 150mm C/C = 1206mm2/metre
FY=250, 4No.16mm BARS AT 250mm C/C = 804mm2/metre

ELEMENT		LONG. REINF (mm2/m)	MOM-X /LOAD (kN-m/m)	TRANS. REINF (mm2/m)	MOM-Y /LOAD (kN-m/m)
2087	TOP :	975.	3.52 / 1112	975.	0.56 / 1112
	BOTT:	975.	-1.47 / 1512	975.	-0.16 / 1512
2088	TOP :	975.	11.33 / 1112	975.	0.32 / 1112
	BOTT:	975.	-4.59 / 1512	975.	-0.15 / 1512
2089	TOP :	975.	18.98 / 1112	975.	2.06 / 1111
	BOTT:	975.	-7.27 / 1512	975.	-0.78 / 1511
2090	TOP :	975.	11.84 / 1112	975.	3.04 / 1111
	BOTT:	975.	-4.93 / 1512	975.	-1.19 / 1511
2091	TOP :	975.	6.12 / 1112	975.	2.38 / 1111
	BOTT:	975.	-1.54 / 1512	975.	-0.58 / 1511
2092	TOP :	975.	6.14 / 1412	975.	2.38 / 1111
	BOTT:	975.	-1.56 / 1212	975.	-0.59 / 1511

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2093	TOP :	975.	11.65 / 1412	975.	2.98 / 1111
	BOTT:	975.	-4.86 / 1212	975.	-1.18 / 1511
2094	TOP :	975.	18.07 / 1412	975.	1.95 / 1111
	BOTT:	975.	-6.81 / 1212	975.	-0.70 / 1511
2095	TOP :	975.	7.44 / 1412	975.	0.25 / 1412
	BOTT:	975.	-3.03 / 1212	975.	-0.17 / 1212
2096	TOP :	975.	1.32 / 1412	975.	0.61 / 1412
	BOTT:	975.	-0.58 / 1212	975.	-0.21 / 1212
2097	TOP :	975.	3.57 / 1112	975.	0.60 / 1411
	BOTT:	975.	-1.52 / 1512	975.	-0.25 / 1211

2098	TOP :	975.	12.76 / 1112	975.	1.45 / 1112
	BOTT:	975.	-5.07 / 1512	975.	-0.68 / 1512
2099	TOP :	975.	27.10 / 1112	975.	4.01 / 1112
	BOTT:	975.	-9.58 / 1512	975.	-1.48 / 1512
2104	TOP :	975.	24.19 / 1412	975.	3.54 / 1412
	BOTT:	975.	-8.56 / 1212	975.	-1.34 / 1212
2105	TOP :	975.	7.84 / 1412	975.	0.67 / 1412
	BOTT:	975.	-3.27 / 1212	975.	-0.50 / 1212
2106	TOP :	975.	1.22 / 1412	975.	0.77 / 1411
	BOTT:	975.	-0.57 / 1212	975.	-0.65 / 1211
2107	TOP :	975.	3.86 / 1112	975.	2.35 / 1112
	BOTT:	975.	-1.64 / 1512	975.	-0.42 / 1512
2108	TOP :	975.	13.83 / 1112	975.	3.62 / 1112
	BOTT:	975.	-5.39 / 1512	975.	-0.83 / 1512
2109	TOP :	975.	27.48 / 1112	975.	5.08 / 1112
	BOTT:	975.	-9.66 / 1512	975.	-1.38 / 1512
2114	TOP :	975.	24.85 / 1412	975.	4.91 / 1412
	BOTT:	975.	-9.05 / 1212	975.	-1.50 / 1212
2115	TOP :	975.	8.59 / 1412	975.	3.23 / 1412
	BOTT:	975.	-3.56 / 1212	975.	-1.08 / 1212
2116	TOP :	975.	1.44 / 1412	975.	2.26 / 1412
	BOTT:	975.	-0.63 / 1212	975.	-0.82 / 1212
2117	TOP :	975.	3.87 / 1112	975.	3.13 / 1112
	BOTT:	975.	-1.70 / 1512	975.	-0.50 / 1512
2118	TOP :	975.	13.83 / 1112	975.	4.55 / 1112
	BOTT:	975.	-5.37 / 1512	975.	-1.04 / 1512
2119	TOP :	975. ✓	27.48 / 1112	975.	5.85 / 1112

Max (along longitudinal direction) (37)

	BOTT:	975.	-9.90 / 1512	975.	-1.72 / 1512

PAINT BOOTH DUCT SUPPORT -- PAGE NO. 26

2124	TOP :	975.	24.97 / 1412	975.	5.92 / 1412
	BOTT:	975.	-9.84 / 1212	975.	-2.13 / 1212
2125	TOP :	975.	8.41 / 1412	975.	4.47 / 1412
	BOTT:	975.	-3.59 / 1212	975.	-1.56 / 1212
2126	TOP :	975.	1.36 / 1412	975.	3.20 / 1412
	BOTT:	975.	-0.65 / 1212	975.	-1.08 / 1212
2127	TOP :	975.	4.03 / 1112	975.	1.76 / 1112
	BOTT:	975.	-1.77 / 1512	975.	-0.18 / 1512
2128	TOP :	975.	12.66 / 1112	975.	3.63 / 1411
	BOTT:	975.	-4.98 / 1512	975.	-1.12 / 1211
2129	TOP :	975.	27.03 / 1112	975.	6.06 / 1112
	BOTT:	975.	-10.10 / 1512	975.	-2.30 / 1512
2134	TOP :	975.	24.66 / 1412	975.	6.11 / 1412
	BOTT:	975.	-10.54 / 1212	975.	-2.76 / 1212
2135	TOP :	975.	7.30 / 1412	975.	3.43 / 1411
	BOTT:	975.	-3.22 / 1212	975.	-1.41 / 1211
2136	TOP :	975.	1.11 / 1412	975.	2.05 / 1411
	BOTT:	975.	-0.54 / 1212	975.	-0.69 / 1211
2137	TOP :	975.	5.47 / 1112	975.	0.95 / 1112
	BOTT:	975.	-2.26 / 1512	975.	-0.22 / 1512
2138	TOP :	975.	10.74 / 1112	975.	2.02 / 1112
	BOTT:	975.	-4.41 / 1512	975.	-0.61 / 1512
2139	TOP :	975.	20.63 / 1112	975.	9.36 / 1411
	BOTT:	975.	-7.78 / 1512	975.	-4.40 / 1211
2140	TOP :	975.	9.72 / 1112	975.	13.71 / 1411
	BOTT:	975.	-3.76 / 1512	975.	-7.36 / 1211
2141	TOP :	975.	5.45 / 1112	975.	12.02 / 1411
	BOTT:	975.	-0.97 / 1512	975.	-6.48 / 1211

PAINT BOOTH DUCT SUPPORT -- PAGE NO. 27

2142	TOP :	975.	5.90 / 1412	975.	12.21 / 1411	
	BOTT:	975.	-1.52 / 1212	975.	-6.92 / 1211	
2143	TOP :	975.	10.34 / 1412	975. ✓	14.17 / 1411	MAX (along
	BOTT:	975.	-4.71 / 1212	975.	-8.43 / 1211	transverse
2144	TOP :	975.	20.21 / 1412	975.	8.95 / 1411	direction
	BOTT:	975.	-8.99 / 1212	975.	-4.70 / 1211	

(38)

2145	TOP :	975.	6.65 / 1412	975.	1.79 / 1412
	BOTT:	975.	-2.96 / 1212	975.	-0.63 / 1212
2146	TOP :	975.	1.07 / 1412	975.	1.62 / 1411
	BOTT:	975.	-0.49 / 1212	975.	-0.62 / 1211
2148	TOP :	975.	6.56 / 1112	975.	2.37 / 1112
	BOTT:	975.	-2.52 / 1512	975.	-1.00 / 1512
2149	TOP :	975.	11.43 / 1112	975.	4.33 / 1411
	BOTT:	975.	-4.28 / 1512	975.	-2.19 / 1211
2150	TOP :	975.	11.49 / 1112	975.	5.89 / 1411
	BOTT:	975.	-4.21 / 1512	975.	-3.35 / 1211
2151	TOP :	975.	6.67 / 1112	975.	5.86 / 1411
	BOTT:	975.	-0.76 / 1512	975.	-3.53 / 1211
2152	TOP :	975.	7.66 / 1412	975.	5.99 / 1411
	BOTT:	975.	-1.84 / 1212	975.	-3.75 / 1211
2153	TOP :	975.	12.95 / 1412	975.	6.24 / 1411
	BOTT:	975.	-5.84 / 1212	975.	-3.94 / 1211
2154	TOP :	975.	13.25 / 1412	975.	4.44 / 1411
	BOTT:	975.	-6.04 / 1212	975.	-2.47 / 1211
2155	TOP :	975.	6.15 / 1412	975.	1.65 / 1412
	BOTT:	975.	-2.76 / 1212	975.	-0.66 / 1212
2156	TOP :	975.	1.07 / 1412	975.	0.92 / 1411
	BOTT:	975.	-0.47 / 1212	975.	-0.30 / 1211

PAINT BOOTH DUCT SUPPORT -- PAGE NO. 28

2181	TOP :	975.	2.70 / 1112	975.	0.54 / 1112
	BOTT:	975.	-0.83 / 1512	975.	-0.31 / 1512
2182	TOP :	975.	7.95 / 1112	975.	0.76 / 1411
	BOTT:	975.	-2.78 / 1512	975.	-0.42 / 1211
2183	TOP :	975.	9.58 / 1112	975.	1.00 / 1411
	BOTT:	975.	-3.21 / 1512	975.	-0.69 / 1211
2184	TOP :	975.	6.94 / 1411	975.	1.17 / 1411
	BOTT:	975.	-0.38 / 1211	975.	-0.78 / 1211
2185	TOP :	975.	8.22 / 1412	975.	1.21 / 1411
	BOTT:	975.	-1.70 / 1212	975.	-0.83 / 1211
2186	TOP :	975.	11.81 / 1412	975.	1.04 / 1411
	BOTT:	975.	-5.22 / 1212	975.	-0.72 / 1211
2187	TOP :	975.	10.79 / 1412	975.	0.88 / 1411
	BOTT:	975.	-5.01 / 1212	975.	-0.55 / 1211
2188	TOP :	975.	5.39 / 1412	975.	0.49 / 1411

(39)

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     BOTT:        975.        -2.49 / 1212        975.        -0.23 / 1211


2189 TOP :        975.         1.06 / 1412        975.         0.40 / 1412
     BOTT:        975.        -0.51 / 1212        975.        -0.18 / 1212


***************************END OF ELEMENT DESIGN**************************

218. END CONCRETE DESIGN
219. FINISH

          *********** END OF THE STAAD.Pro RUN ***********

            **** DATE= MAY 12,2009   TIME= 22:12:12 ****
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SPRING REACTIONS

Node	Area	L/C	Force-Y kN	L/C	Force-Y kN	L/C	Force-Y kN	L/C	Force-Y kN	L/C	Force-Y kN
11	0.16	2001	4.879	2011	6.722	2012	5.925	2111	3.036	2112	3.833
12	0.16	2001	4.915	2011	6.83	2012	3.822	2111	2.999	2112	6.007
13	0.161	2001	4.293	2011	3.373	2012	3.311	2111	5.213	2112	5.276
14	0.161	2001	4.257	2011	3.264	2012	5.424	2111	5.251	2112	3.091
70	0.047	2001	1.523	2011	2.453	2012	2.587	2111	0.592	2112	0.458
71	0.078	2001	2.504	2011	4.048	2012	3.76	2111	0.961	2112	1.249
72	0.056	2001	1.815	2011	2.939	2012	2.498	2111	0.691	2112	1.131
73	0.045	2001	1.469	2011	2.383	2012	1.866	2111	0.555	2112	1.073
74	0.069	2001	2.243	2011	3.642	2012	2.667	2111	0.843	2112	1.818
75	0.099	2001	3.236	2011	5.271	2012	3.186	2111	1.201	2112	3.286
76	0.069	2001	2.258	2011	3.689	2012	1.763	2111	0.827	2112	2.752
77	0.057	2001	1.879	2011	3.073	2012	1.319	2111	0.685	2112	2.439
78	0.075	2001	2.462	2011	4.034	2012	1.351	2111	0.889	2112	3.572
79	0.066	2001	2.153	2011	3.534	2012	0.862	2111	0.771	2112	3.443
80	0.028	2001	0.935	2011	1.538	2012	0.268	2111	0.332	2112	1.603
81	0.078	2001	2.42	2011	3.543	2012	4.195	2111	1.298	2112	0.645
82	0.128	2001	3.982	2011	5.851	2012	6.084	2111	2.112	2112	1.879
83	0.093	2001	2.885	2011	4.251	2012	4.034	2111	1.52	2112	1.736
84	0.075	2001	2.337	2011	3.45	2012	3.008	2111	1.223	2112	1.665
85	0.114	2001	3.567	2011	5.274	2012	4.294	2111	1.859	2112	2.839
86	0.164	2001	5.146	2011	7.638	2012	5.099	2111	2.655	2112	5.194
87	0.114	2001	3.592	2011	5.351	2012	2.799	2111	1.833	2112	4.385
88	0.095	2001	2.989	2011	4.459	2012	2.084	2111	1.519	2112	3.894
89	0.124	2001	3.916	2011	5.856	2012	2.109	2111	1.976	2112	5.723
90	0.108	2001	3.425	2011	5.134	2012	1.316	2111	1.716	2112	5.533
91	0.047	2001	1.488	2011	2.236	2012	0.396	2111	0.741	2112	2.581
92	0.11	2001	3.31	2011	4.505	2012	5.816	2111	2.114	2112	0.803
93	0.18	2001	5.446	2011	7.447	2012	8.421	2111	3.445	2112	2.47
94	0.13	2001	3.947	2011	5.413	2012	5.578	2111	2.481	2112	2.316
95	0.105	2001	3.196	2011	4.395	2012	4.154	2111	1.998	2112	2.239
97	0.23	2001	7.04	2011	9.738	2012	7.006	2111	4.342	2112	7.074
99	0.133	2001	4.09	2011	5.692	2012	2.838	2111	2.488	2112	5.341
100	0.174	2001	5.358	2011	7.479	2012	2.847	2111	3.238	2112	7.87
101	0.152	2001	4.686	2011	6.56	2012	1.749	2111	2.813	2112	7.624
102	0.066	2001	2.037	2011	2.858	2012	0.513	2111	1.216	2112	3.56
103	0.157	2001	4.418	2011	4.741	2012	8.061	2111	4.095	2112	0.774
104	0.258	2001	7.271	2011	7.858	2012	11.622	2111	6.684	2112	2.921
105	0.186	2001	5.271	2011	5.722	2012	7.671	2111	4.82	2112	2.871
106	0.15	2001	4.269	2011	4.652	2012	5.692	2111	3.887	2112	2.847
108	0.19	2001	5.467	2011	6.071	2012	3.747	2111	4.863	2112	7.187
109	0.249	2001	7.164	2011	7.997	2012	3.661	2111	6.332	2112	10.668
110	0.218	2001	6.267	2011	7.029	2012	2.142	2111	5.504	2112	10.391
111	0.094	2001	2.724	2011	3.067	2012	0.579	2111	2.381	2112	4.869
112	0.111	2001	2.882	2011	2.143	2012	5.482	2111	3.622	2112	0.283
113	0.181	2001	4.746	2011	3.573	2012	7.868	2111	5.92	2112	1.625
114	0.131	2001	3.442	2011	2.611	2012	5.175	2111	4.272	2112	1.708
115	0.106	2001	2.788	2011	2.128	2012	3.826	2111	3.449	2112	1.75
117	0.134	2001	3.574	2011	2.822	2012	2.412	2111	4.325	2112	4.735
118	0.175	2001	4.684	2011	3.735	2012	2.286	2111	5.632	2112	7.082
119	0.153	2001	4.098	2011	3.298	2012	1.256	2111	4.898	2112	6.94
120	0.066	2001	1.781	2011	1.443	2012	0.298	2111	2.119	2112	3.264
121	0.053	2001	1.334	2011	0.79	2012	2.585	2111	1.879	2112	0.084
122	0.087	2001	2.198	2011	1.324	2012	3.703	2111	3.071	2112	0.693
123	0.063	2001	1.594	2011	0.97	2012	2.431	2111	2.217	2112	0.756
124	0.051	2001	1.291	2011	0.793	2012	1.795	2111	1.79	2112	0.788
125	0.077	2001	1.972	2011	1.219	2012	2.54	2111	2.725	2112	1.404
126	0.111	2001	2.848	2011	1.79	2012	2.927	2111	3.905	2112	2.768
127	0.077	2001	1.989	2011	1.271	2012	1.532	2111	2.707	2112	2.446
128	0.064	2001	1.656	2011	1.065	2012	1.11	2111	2.246	2112	2.201
129	0.084	2001	2.17	2011	1.415	2012	1.037	2111	2.926	2112	3.304

SPRING REACTIONS

Node	Area	L/C	Force-Y kN	L/C	Force-Y kN	L/C	Force-Y kN	L/C	Force-Y kN	L/C	Force-Y kN
130	0.073	2001	1.899	2011	1.253	2012	0.551	2111	2.545	2112	3.246
131	0.032	2001	0.826	2011	0.55	2012	0.121	2111	1.101	2112	1.53
132	0.021	2001	1.432	2011	0.701	2012	2.808	2111	2.162	2112	0.056
133	0.06	2001	2.515	2011	1.261	2012	4.283	2111	3.769	2112	0.747
134	0.073	2001	1.824	2011	0.927	2012	2.809	2111	2.72	2112	0.838
135	0.059	2001	1.478	2011	0.759	2012	2.071	2111	2.196	2112	0.884
136	0.09	2001	2.256	2011	1.169	2012	2.928	2111	3.344	2112	1.585
137	0.129	2001	3.259	2011	1.725	2012	3.364	2111	4.793	2112	3.154
138	0.09	2001	2.277	2011	1.231	2012	1.752	2111	3.323	2112	2.801
139	0.075	2001	1.895	2011	1.033	2012	1.265	2111	2.757	2112	2.525
140	0.098	2001	2.484	2011	1.376	2012	1.169	2111	3.593	2112	3.799
141	0.085	2001	2.174	2011	1.223	2012	0.607	2111	3.125	2112	3.74
142	0.037	2001	0.945	2011	0.538	2012	0.126	2111	1.353	2112	1.765
144	0.053	2001	3.055	2011	0.872	2012	5.324	2111	5.239	2112	0.787
145	0.097	2001	2.327	2011	0.683	2012	3.658	2111	3.972	2112	0.996
146	0.079	2001	1.886	2011	0.566	2012	2.69	2111	3.206	2112	1.082
147	0.12	2001	2.88	2011	0.879	2012	3.796	2111	4.882	2112	1.965
148	0.172	2001	4.162	2011	1.322	2012	4.334	2111	7.001	2112	3.989
149	0.12	2001	2.907	2011	0.961	2012	2.232	2111	4.854	2112	3.582
150	0.1	2001	2.42	2011	0.812	2012	1.601	2111	4.028	2112	3.239
151	0.13	2001	3.173	2011	1.096	2012	1.447	2111	5.251	2112	4.9
152	0.114	2001	2.777	2011	0.985	2012	0.712	2111	4.569	2112	4.843
153	0.05	2001	1.208	2011	0.437	2012	0.125	2111	1.978	2112	2.29
154	0.229	2001	6.568	2011	7.275	2012	5.094	2111	5.861	2112	8.042
155	0.329	2001	9.407	2011	10.349	2012	9.477	2111	8.465	2112	9.337
156	0.229	2001	6.517	2011	7.12	2012	8.089	2111	5.914	2112	4.945
157	0.231	2001	6.147	2011	4.775	2012	6.279	2111	7.518	2112	6.014
158	0.026	2001	0.83	2011	0.041	2012	1.482	2111	1.62	2112	0.178
160	0.049	2001	1.112	2011	0.065	2012	1.786	2111	2.158	2112	0.438
161	0.039	2001	0.901	2011	0.06	2012	1.31	2111	1.742	2112	0.492
162	0.06	2001	1.376	2011	0.099	2012	1.845	2111	2.653	2112	0.908
163	0.086	2001	1.989	2011	0.172	2012	2.091	2111	3.806	2112	1.886
164	0.06	2001	1.39	2011	0.14	2012	1.064	2111	2.639	2112	1.715
165	0.05	2001	1.157	2011	0.123	2012	0.758	2111	2.19	2112	1.556
166	0.065	2001	1.517	2011	0.178	2012	0.667	2111	2.856	2112	2.367
167	0.057	2001	1.328	2011	0.17	2012	0.307	2111	2.486	2112	2.349
168	0.025	2001	0.577	2011	0.079	2012	0.041	2111	1.076	2112	1.114

SOIL PRESSURES (kPa)

Node	L/C	Soil Pressure	L/C	Soil Pressure	L/C	Soil Pressure	L/C	Soil Pressure	L/C	Soil Pressure
11	2001	30.5	2011	42.0	2012	37.0	2111	19.0	2112	24.0
12	2001	4.915	2011	6.83	2012	3.822	2111	2.999	2112	6.007
13	2001	4.293	2011	3.373	2012	3.311	2111	5.213	2112	5.276
14	2001	4.257	2011	3.264	2012	5.424	2111	5.251	2112	3.091
70	2001	1.523	2011	2.453	2012	2.587	2111	0.592	2112	0.458
71	2001	2.504	2011	4.048	2012	3.76	2111	0.961	2112	1.249
72	2001	1.815	2011	2.939	2012	2.498	2111	0.691	2112	1.131
73	2001	1.469	2011	2.383	2012	1.866	2111	0.555	2112	1.073
74	2001	2.243	2011	3.642	2012	2.667	2111	0.843	2112	1.818
75	2001	3.236	2011	5.271	2012	3.186	2111	1.201	2112	3.286
76	2001	2.258	2011	3.689	2012	1.763	2111	0.827	2112	2.752
77	2001	1.879	2011	3.073	2012	1.319	2111	0.685	2112	2.439
78	2001	2.462	2011	4.034	2012	1.351	2111	0.889	2112	3.572
79	2001	2.153	2011	3.534	2012	0.862	2111	0.771	2112	3.443
80	2001	0.935	2011	1.538	2012	0.268	2111	0.332	2112	1.603
81	2001	2.42	2011	3.543	2012	4.195	2111	1.298	2112	0.645
82	2001	3.982	2011	5.851	2012	6.084	2111	2.112	2112	1.879
83	2001	2.885	2011	4.251	2012	4.034	2111	1.52	2112	1.736
84	2001	2.337	2011	3.45	2012	3.008	2111	1.223	2112	1.665
85	2001	3.567	2011	5.274	2012	4.294	2111	1.859	2112	2.839
86	2001	5.146	2011	7.638	2012	5.099	2111	2.655	2112	5.194
87	2001	3.592	2011	5.351	2012	2.799	2111	1.833	2112	4.385
88	2001	2.989	2011	4.459	2012	2.084	2111	1.519	2112	3.894
89	2001	3.916	2011	5.856	2012	2.109	2111	1.976	2112	5.723
90	2001	3.425	2011	5.134	2012	1.316	2111	1.716	2112	5.533
91	2001	1.488	2011	2.236	2012	0.396	2111	0.741	2112	2.581
92	2001	3.31	2011	4.505	2012	5.816	2111	2.114	2112	0.803
93	2001	5.446	2011	7.447	2012	8.421	2111	3.445	2112	2.47
94	2001	3.947	2011	5.413	2012	5.578	2111	2.481	2112	2.316
95	2001	3.196	2011	4.395	2012	4.154	2111	1.998	2112	2.239
97	2001	7.04	2011	9.738	2012	7.006	2111	4.342	2112	7.074
99	2001	4.09	2011	5.692	2012	2.838	2111	2.488	2112	5.341
100	2001	5.358	2011	7.479	2012	2.847	2111	3.238	2112	7.87
101	2001	4.686	2011	6.56	2012	1.749	2111	2.813	2112	7.624
102	2001	2.037	2011	2.858	2012	0.513	2111	1.216	2112	3.56
103	2001	4.418	2011	4.741	2012	8.061	2111	4.095	2112	0.774
104	2001	7.271	2011	7.858	2012	11.622	2111	6.684	2112	2.921
105	2001	5.271	2011	5.722	2012	7.671	2111	4.82	2112	2.871
106	2001	4.269	2011	4.652	2012	5.692	2111	3.887	2112	2.847
108	2001	5.467	2011	6.071	2012	3.747	2111	4.863	2112	7.187
109	2001	7.164	2011	7.997	2012	3.661	2111	6.332	2112	10.668
110	2001	6.267	2011	7.029	2012	2.142	2111	5.504	2112	10.391
111	2001	2.724	2011	3.067	2012	0.579	2111	2.381	2112	4.869
112	2001	2.882	2011	2.143	2012	5.482	2111	3.622	2112	0.283
113	2001	4.746	2011	3.573	2012	7.868	2111	5.92	2112	1.625
114	2001	3.442	2011	2.611	2012	5.175	2111	4.272	2112	1.708
115	2001	2.788	2011	2.128	2012	3.826	2111	3.449	2112	1.75
117	2001	3.574	2011	2.822	2012	2.412	2111	4.325	2112	4.735
118	2001	4.684	2011	3.735	2012	2.286	2111	5.632	2112	7.082
119	2001	4.098	2011	3.298	2012	1.256	2111	4.898	2112	6.94
120	2001	1.781	2011	1.443	2012	0.298	2111	2.119	2112	3.264
121	2001	1.334	2011	0.79	2012	2.585	2111	1.879	2112	0.084
122	2001	2.198	2011	1.324	2012	3.703	2111	3.071	2112	0.693
123	2001	1.594	2011	0.97	2012	2.431	2111	2.217	2112	0.756
124	2001	1.291	2011	0.793	2012	1.795	2111	1.79	2112	0.788
125	2001	1.972	2011	1.219	2012	2.54	2111	2.725	2112	1.404
126	2001	2.848	2011	1.79	2012	2.927	2111	3.905	2112	2.768
127	2001	1.989	2011	1.271	2012	1.532	2111	2.707	2112	2.446
128	2001	1.656	2011	1.065	2012	1.11	2111	2.246	2112	2.201
129	2001	2.17	2011	1.415	2012	1.037	2111	2.926	2112	3.304
130	2001	1.899	2011	1.253	2012	0.551	2111	2.545	2112	3.246
131	2001	0.826	2011	0.55	2012	0.121	2111	1.101	2112	1.53
132	2001	1.432	2011	0.701	2012	2.808	2111	2.162	2112	0.056
133	2001	2.515	2011	1.261	2012	4.283	2111	3.769	2112	0.747
134	2001	1.824	2011	0.927	2012	2.809	2111	2.72	2112	0.838
135	2001	1.478	2011	0.759	2012	2.071	2111	2.196	2112	0.884
136	2001	2.256	2011	1.169	2012	2.928	2111	3.344	2112	1.585
137	2001	3.259	2011	1.725	2012	3.364	2111	4.793	2112	3.154
138	2001	2.277	2011	1.231	2012	1.752	2111	3.323	2112	2.801
139	2001	1.895	2011	1.033	2012	1.265	2111	2.757	2112	2.525
140	2001	2.484	2011	1.376	2012	1.169	2111	3.593	2112	3.799
141	2001	2.174	2011	1.223	2012	0.607	2111	3.125	2112	3.74
142	2001	0.945	2011	0.538	2012	0.126	2111	1.353	2112	1.765
144	2001	3.055	2011	0.872	2012	5.324	2111	5.239	2112	0.787
145	2001	2.327	2011	0.683	2012	3.658	2111	3.972	2112	0.996
146	2001	1.886	2011	0.566	2012	2.69	2111	3.206	2112	1.082

SOIL PRESSURES (kPa)

Node	L/C	Soil Pressure	L/C	Soil Pressure	L/C	Soil Pressure	L/C	Soil Pressure	L/C	Soil Pressure
147	2001	2.88	2011	0.879	2012	3.796	2111	4.882	2112	1.965
148	2001	4.162	2011	1.322	2012	4.334	2111	7.001	2112	3.989
149	2001	2.907	2011	0.961	2012	2.232	2111	4.854	2112	3.582
150	2001	2.42	2011	0.812	2012	1.601	2111	4.028	2112	3.239
151	2001	3.173	2011	1.096	2012	1.447	2111	5.251	2112	4.9
152	2001	2.777	2011	0.985	2012	0.712	2111	4.569	2112	4.843
153	2001	1.208	2011	0.437	2012	0.125	2111	1.978	2112	2.29
154	2001	6.568	2011	7.275	2012	5.094	2111	5.861	2112	8.042
155	2001	9.407	2011	10.349	2012	9.477	2111	8.465	2112	9.337
156	2001	6.517	2011	7.12	2012	8.089	2111	5.914	2112	4.945
157	2001	6.147	2011	4.775	2012	6.279	2111	7.518	2112	6.014
158	2001	0.83	2011	0.041	2012	1.482	2111	1.62	2112	0.178
160	2001	1.112	2011	0.065	2012	1.786	2111	2.158	2112	0.438
161	2001	0.901	2011	0.06	2012	1.31	2111	1.742	2112	0.492
162	2001	1.376	2011	0.099	2012	1.845	2111	2.653	2112	0.908
163	2001	1.989	2011	0.172	2012	2.091	2111	3.806	2112	1.886
164	2001	1.39	2011	0.14	2012	1.064	2111	2.639	2112	1.715
165	2001	1.157	2011	0.123	2012	0.758	2111	2.19	2112	1.556
166	2001	1.517	2011	0.178	2012	0.667	2111	2.856	2112	2.367
167	2001	1.328	2011	0.17	2012	0.307	2111	2.486	2112	2.349
168	2001	0.577	2011	0.079	2012	0.041	2111	1.076	2112	1.114
Max soil pressure:		30.5		42.0		37.0		19.0		24.0
Min soil pressure:		0.58		0.04		0.04		0.33		0.06
(Allo. = 150 kPa)										

SUPPORT FRAME REACTION

Node	L/C	Force-X kN	Force-Y kN	Force-Z kN
11	1112	0.602	118.579	1.258
12	1412	-18.926	118.123	0.889
11	1212	0.561	114.631	1.219
12	1512	-18.888	114.306	0.853
13	1412	0.121	106.927	-0.629
14	1112	19.093 ✓	106.471	-0.98
13	1512	0.159	103.042	-0.59
14	1212	19.058	102.717	-0.943
11	1012	0.516	101.639	1.078
12	1312	-16.222	101.249	0.762
13	1312	0.103	91.651	-0.539
14	1012	16.366	91.261	-0.84
13	1411	-0.624	76.78	-11.797
12	1111	-0.685	76.545	0.43
11	1111	0.657	75.641	12.298 ✓
14	1411	0.578	74.965	0.142
13	1511	-0.586	72.896	-11.758
12	1211	-0.647	72.728	0.393
11	1211	0.617	71.693	12.259
14	1511	0.543	71.211	0.179
13	1311	-0.535	65.812	-10.112
12	1011	-0.587	65.61	0.368
11	1011	0.563	64.835	10.541
14	1311	0.496	64.256	0.122
11	1022	-0.022	41.597	0.434
12	1122	-5.129	41.207	0.328
13	1121	-0.339	40.268	-5.068
12	1021	-0.364	40.067	-0.029
11	1021	0.358	39.821	5.035
14	1121	0.317	39.242	0.023
11	1222	-0.052	38.637	0.405
12	1322	-5.1	38.344	0.301
13	1122	-0.031	38.246	-0.265
14	1022	4.919	37.856	-0.355
13	1321	-0.311	37.355	-5.039
12	1221	-0.336	37.204	-0.056
11	1221	0.327	36.861	5.006
14	1321	0.29	36.426	0.05
13	1322	-0.002	35.333	-0.236
14	1222	4.892	35.04	-0.327
11	1001	0.141	13.816	0.137
13	1001	-0.134	13.595	-0.135
12	1001	-0.132	13.361	0.126
14	1001	0.125	13.14	-0.128
13	1022	-0.198	-14.94	0.033
14	1122	-4.705	-15.33	0.136
11	1121	-0.115	-16.137	-4.801
14	1021	-0.103	-16.716	-0.242
13	1021	0.11	-16.962	4.836
12	1121	0.138	-17.163	0.246
13	1222	-0.17	-17.853	0.062
11	1122	0.265	-17.913	-0.199
14	1322	-4.732	-18.146	0.163
12	1022	4.902	-18.303	-0.112
11	1321	-0.146	-19.097	-4.83
14	1221	-0.13	-19.532	-0.215
13	1221	0.139	-19.875	4.865
12	1321	0.166	-20.026	0.219
11	1322	0.234	-20.873	-0.229
12	1222	4.93	-21.166	-0.139
11	1311	-0.321	-41.151	-10.306
14	1011	-0.282	-41.73	-0.341
13	1011	0.306	-42.505	9.88
12	1311	0.36	-42.706	-0.152
11	1411	-0.374	-48.009	-12.024
14	1111	-0.329	-48.685	-0.398
13	1111	0.357	-49.589	11.526
12	1411	0.42	-49.824	-0.177
11	1511	-0.415	-51.956	-12.063
14	1211	-0.365	-52.439	-0.361
13	1211	0.395	-53.474	11.565
12	1511	0.458	-53.641	-0.213
13	1012	-0.333	-68.345	0.307
14	1312	-16.152	-68.735	0.621
11	1312	-0.273	-77.954	-0.844
12	1012	15.995	-78.345	-0.545
13	1112	-0.388	-79.736	0.358
14	1412	-18.844	-80.191	0.724
13	1212	-0.35	-83.62	0.397
14	1512	-18.88	-83.945	0.761
11	1412	-0.319	-90.947	-0.984
12	1112	18.661	-91.402	-0.636
11	1512	-0.359	-94.894	-1.023
12	1212	18.699	-95.219 ✓	-0.672

FX MAX

FZ max

MAX UPLIFT

- Check for bot tension;

Assuming 50% of bolt will carry tension only;

$T_{per\ bolt} = \frac{95}{2} = 48\ kN < 110\ kN$

(from p. 45)

- check for shear transfer,

Assuming 50% of bolt will carry shear only;

$V_{per\ bolt} = \frac{19}{2} = 9.5\ kN$

$V_{cap} = 39\ kN > 9.5$ (p. 46)

ok!

BOLT CAPACITIES

BS 5950-1: 2000
BS 4190: 2001

Table H.49 Non-Preloaded Ordinary Bolts in S275

GRADE 4.6 BOLTS IN S275

Diameter of Bolt	Tensile Stress Area	Tension Capacity		Shear Capacity		Bearing Capacity in kN (Minimum of P_{bb} and P_{bs}) End distance equal to 2 x bolt diameter. Thickness in mm of ply passed through.										
		Nominal $0.8A_tp_t$	Exact A_tp_t	Single Shear	Double Shear											
	A_t	P_{nom}	P_t	P_s	$2P_s$											
mm	mm^2	kN	kN	kN	kN	5	6	7	8	9	10	12	15	20	25	30
12	84.3	16.2	20.2	13.5	27.0	*27.6*	*33.1*	*38.6*	*44.2*	*49.7*	*55.2*	*66.2*	*82.8*	*110*	*138*	*166*
16	157	30.1	37.7	25.1	50.2	36.8	44.2	*51.5*	*58.9*	*66.2*	*73.6*	*88.3*	*110*	*147*	*184*	*221*
20	245	47.0	58.8	39.2	78.4	46.0	55.2	64.4	73.6	*82.8*	*92.0*	*110*	*138*	*184*	*230*	*276*
22	303	58.2	72.7	48.5	97.0	50.6	60.7	70.8	81.0	91.1	*101*	*121*	*152*	*202*	*253*	*304*
24	353	67.8	84.7	56.5	113	**55.2**	66.2	77.3	88.3	99.4	110	*132*	*166*	*221*	*276*	*331*
27	459	88.1	110	73.4	147	**62.1**	74.5	86.9	99.4	112	124	*149*	*186*	*248*	*311*	*373*
30	561	108	135	89.8	180	**69.0**	**82.8**	96.6	110	124	138	166	*207*	*276*	*345*	*414*

See notes below

GRADE 8.8 BOLTS IN S275

Diameter of Bolt	Tensile Stress Area	Tension Capacity		Shear Capacity		Bearing Capacity in kN (Minimum of P_{bb} and P_{bs}) End distance equal to 2 x bolt diameter. Thickness in mm of ply passed through.										
		Nominal $0.8A_tp_t$	Exact A_tp_t	Single Shear	Double Shear											
	A_t	P_{nom}	P_t	P_s	$2P_s$											
mm	mm^2	kN	kN	kN	kN	5	6	7	8	9	10	12	15	20	25	30
12	84.3	37.8	47.2	31.6	63.2	**27.6**	33.1	38.6	44.2	49.7	55.2	*66.2*	*82.8*	*110*	*138*	*166*
16	157	70.3	87.9	58.9	118	**36.8**	**44.2**	**51.5**	58.9	66.2	73.6	88.3	110	*147*	*184*	*221*
20	245	110	137	91.9	184	**46.0**	**55.2**	**64.4**	**73.6**	**82.8**	92.0	110	138	*184*	*230*	*276*
22	303	136	170	114	227	**50.6**	**60.7**	**70.8**	**81.0**	**91.1**	**101**	121	152	202	*253*	*304*
24	353	158	198	132	265	**55.2**	**66.2**	**77.3**	**88.3**	**99.4**	**110**	132	166	221	276	*331*
27	459	206	257	172	344	**62.1**	**74.5**	**86.9**	**99.4**	**112**	**124**	**149**	186	248	311	*373*
30	561	251	314	210	421	**69.0**	**82.8**	**96.6**	**110**	**124**	**138**	**166**	**207**	276	345	414

See notes below

GRADE 10.9 BOLTS IN S275

Diameter of Bolt	Tensile Stress Area	Tension Capacity		Shear Capacity		Bearing Capacity in kN (Minimum of P_{bb} and P_{bs}) End distance equal to 2 x bolt diameter. Thickness in mm of ply passed through.										
		Nominal $0.8A_tp_t$	Exact A_tp_t	Single Shear	Double Shear											
	A_t	P_{nom}	P_t	P_s	$2P_s$											
mm	mm^2	kN	kN	kN	kN	5	6	7	8	9	10	12	15	20	25	30
12	84.3	47.2	59.0	33.7	67.4	**27.6**	**33.1**	38.6	44.2	49.7	55.2	66.2	*82.8*	*110*	*138*	*166*
16	157	87.9	110	62.8	126	**36.8**	**44.2**	**51.5**	**58.9**	66.2	73.6	88.3	110	*147*	*184*	*221*
20	245	137	172	98.0	196	**46.0**	**55.2**	**64.4**	**73.6**	**82.8**	**92.0**	110	138	184	*230*	*276*
22	303	170	212	121	242	**50.6**	**60.7**	**70.8**	**81.0**	**91.1**	**101**	121	152	202	*253*	*304*
24	353	198	247	141	282	**55.2**	**66.2**	**77.3**	**88.3**	**99.4**	**110**	**132**	166	221	276	*331*
27	459	257	321	184	367	**62.1**	**74.5**	**86.9**	**99.4**	**112**	**124**	**149**	186	248	311	*373*
30	561	314	393	224	449	**69.0**	**82.8**	**96.6**	**110**	**124**	**138**	**166**	**207**	276	345	414

Values in **bold** are less than the single shear capacity of the bolt.

Values in *italic* are greater than the double shear capacity of the bolt.

Bearing values assume standard clearance holes.

If oversize or short slotted holes are used, bearing values should be multiplied by 0.7.

If long slotted or kidney shaped holes are used, bearing values should be multiplied by 0.5.

If appropriate, shear capacity must be reduced for large packings, large grip lengths and long joints.

(46)

HIT-HY 150 injection mortar with HAS rod

HAS-HY 150 WITH HAS, HAS-E

Recommended load, L_{rec} [kN]: concrete $f_{ck,cube}$ = 25 N/mm^2

Anchor size	M8	M10	M12	M16	M20	M24
Tensile, N_{Rec}	6.0	8.0	12.0	15.3	26.0	32.4
Shear, V_{Rec}	5.6	9.0	13.1	24.7	38.6	55.6

Basic loading data (for a single anchor): HIT-HY 150 with HAS-R, HAS-E-R, HAS-HCR

All data on this page applies to

- concrete: See table below.
- correct setting (See setting operations page 260)
- no edge distance and spacing influence
- steel failure

For detailed design method, see pages 261 – 265.

CONC non-cracked concrete

Mean ultimate resistance, $R_{u,m}$ [kN]: concrete ≅ C20/25

Anchor size	M8	M10	M12	M16	M20	M24
Tensile, $N_{Ru,m}$	24.8	39.6	57.8	109.1	170.3	244.4
Shear, $V_{Ru,m}$	14.8	23.8	34.5	65.4	102.1	146.9

Characteristic resistance, R_k [kN]: concrete ≅ C20/25

Anchor size	M8	M10	M12	M16	M20	M24
Tensile, N_{Rk}	23.0	36.7	53.5	101.0	157.6	226.3
Shear, V_{Rk}	13.7	22.0	32.0	60.5	94.5	136.0

Following values according to the

Concrete Capacity Method

Design resistance, R_d [kN]: concrete $f_{ck,cube}$ = 25 N/mm^2

Anchor size	M8	M10	M12	M16	M20	M24
Tensile, N_{Rd}	8.4	11.2	16.8	21.4	36.4	45.4
Shear, V_{Rd}	8.8	14.1	20.5	38.8	60.6	87.2

Recommended load, L_{rec} [kN]: concrete $f_{ck,cube}$ = 25 N/mm^2

Anchor size	M8	M10	M12	M16	M20	M24
Tensile, N_{Rec}	6.0	8.0	12.0	15.3	26.0	32.4
Shear, V_{Rec}	6.3	10.1	14.6	27.7	43.3	62.3

(47)

3. AHU Platform – Lathe on Pit

The following is a list of drawings associated with this submission:

Project:	Dubai Metro, UAE - I Rashidiya , Building 3			ATKINS	
Structure:	AHU Support Platform				
Calculations:					
By	Abhijeet	**Date**	Apr'2009	**Sheet**	
Ref:	**Calculations**				**Output**

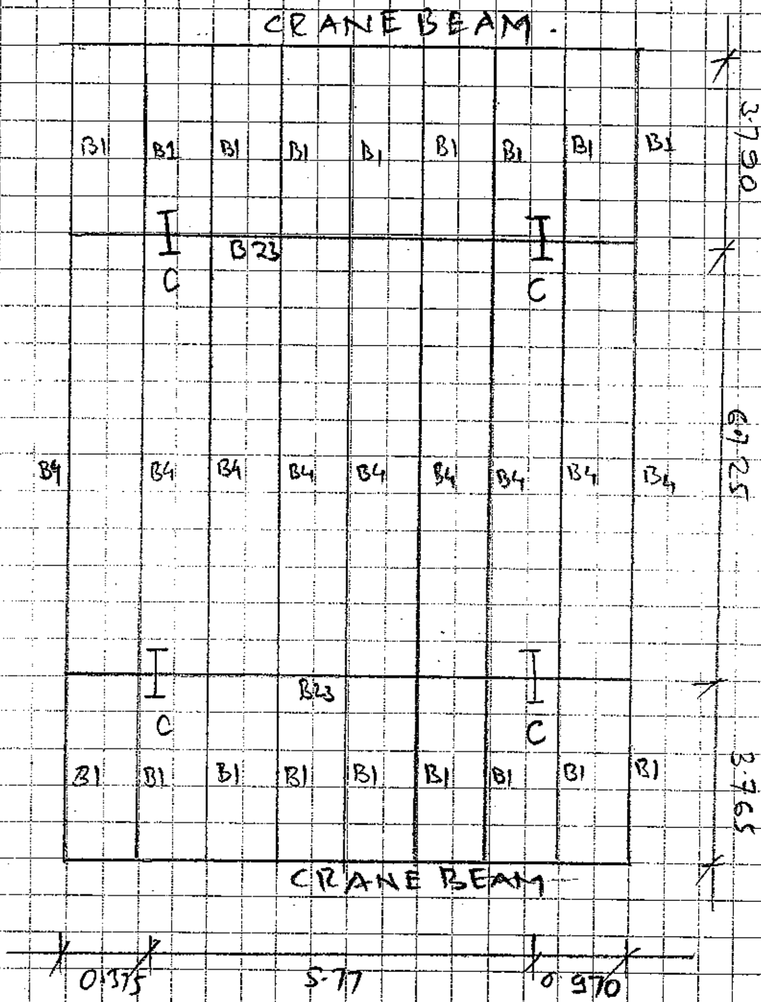
Calculation Index

Section Sheet No

ATKINS
A member of the WS Atkins Group
PO Box 5620
Dubai, United Arab Emirates

Project	DUBAI METRO			Job ref.
Part of Structure	AHU SUPPORT PLATFORM			Calc. sheet no. 2 / / rev.
Drawing ref.		Calc. by Abhijeet	Date Apr'2009	Check by / Date

Ref.	Calculations	Output



Load Path

AHU PLATFORM ⟶ BEAM B4, BEAM B1

COLUMN ⟵ BEAM B-23

(50)

ATKINS A member of the WS Atkins Group PO Box 5620 Dubai, United Arab Emirates	Project DUBAI METRO				Job ref.	
	Part of Structure AHU SUPPORT PLATFORM				Calc. sheet no. 3 /	rev. /
	Drawing ref.	Calc. by Abhijeet	Date APR' 2009		Check by	Date

Ref.	Calculations	Output

LOAD CALCULATION :-

DL

1. Chequered plate = 0.4 KN/m²
2. Mechanical and Electrical Equipment = 4.17 KN/m²
3. Services = 0.5 KN/m²
4. Self wt as per Beam Sizes = S.W.B

Σ = 5.07 KN/m²

γ_f = 1.4 = 1.4(5.07)

30% wt for Connection = 1.3(5.07)

12.675 KN/m²

Equip. wt

$\frac{15.50}{2.6 \times 1.4}$ = 4.17 KN/m²

Plan area of Equipment

LL.

As Per (DM001-E-MAD-CDP-DR-DCC-371920-A1)

1. Live Load = 1.5 KN/m²

γ_f = 1.6 ∴ LL = 2.4 KN/m²

Hence Total Surface wt = 15.395

= 15.4 KN/m².

Beam B1

Simply Supported to Beam B.23 and crane beam.

Connection Type – Shear Connection

l_e Effective Span = 3.790 m.

w Udl for beam B.1 = 15.4 × 0.9 + 0.11

= 13.97 KN/m.

factored self wt of Beam (51)

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Project	DUBAI METRO			Job ref.	
Part of Structure	AHU SUPPOR PLATFORM			Calc. sheet no. 4 /	rev. /
Drawing ref.		Calc. by Abhijeet	Date Apr'2009	Check by	Date

Ref.	Calculations	Output

$M_{B1,max}$ Max Bending Moment $= \frac{13.97 \times 3.79^2}{8}$

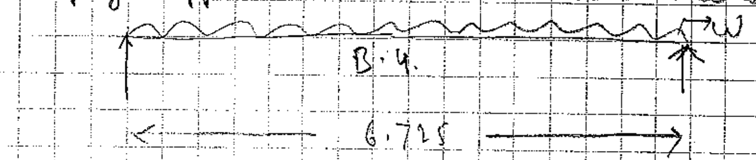
$= 125.08$ KNm.

$V_{B1,Max}$ Max Shear Force $= \frac{52.95 \text{ KN}}{2}$

$= 26.475$ KN.

BEAM B4

Simply Supported Beam - Shear Connection at End.



W → uniformly distributed load $= 15.4 \times 0.9 + 0.589$

$= 14.45$ KN/m

L_e Effective Span $= 6.725$m

$M_{B4,Max}$ Max Bending Moment $= \frac{14.45 \times 6.725^2}{8}$ $= 81.62$ KNm

$V_{B4,max}$ Max Shear Force $= 48.6$ KN.

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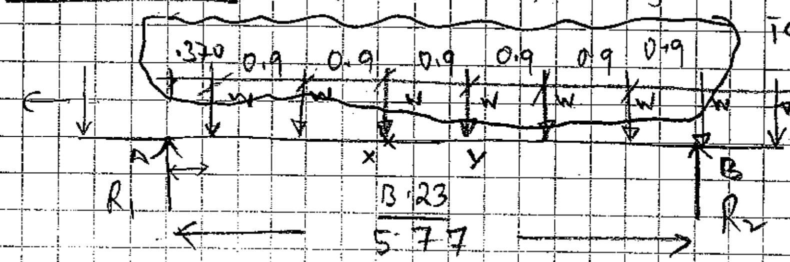
Project DUBAI METRO			Job ref.	
Part of Structure AHU SOPPOR PLATFORM			Calc. sheet no. 5 / /	rev.
Drawing ref.	Calc. by Abhijeet	Date Apr '2009	Check by	Date

Ref.	Calculations	Output

Beam B.23 (Load Transferred from - B4, B1, Taken Load at points (conservatively)

Neglecting this force contribution



Neglecting this Force Contribution

W Point Load Transferred from B1 and B4 = 52.95 + 48.6 = 101.55 KN

Le Effective Span = 5.77 m

Reaction Calculation:

$\Sigma M_A = 0$

$\rightarrow R_2 \times 5.77 = 101.55 \times [4.87 + 3.97 + 3.07 + 2.17 + 1.27 + 0.37]$

$\rightarrow R_2 = 277$ KN

$\therefore R_1 = 333$ KN

M_x Moment at X $= 333 \times 2.17 - 101.55 \times (.37 + 1.8 + 0.9)$
$= 448$ KNm

M_y Moment at Y $= 333 \times 3.07 - 101.55 (2.7 + 1.8 + 0.9)$
$= 1022.31 - 548.37$
$= 474$ KNm.

$\therefore M_{max} = Max(M_x, M_y)$

$\therefore M_{max} = 474$ KNm

$V_{max} = 333$ KN.

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	Part of Structure AHU SUPPORT PLATFORM			Calc. sheet no. 6 /	rev. /
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Ref.	Calculations	Output

Lateral Load Calculation :-

S. No		Length	Number	Weight KN/m	Weight KN
1	Beam B1	7.56	9	0.34	23.14
2	B4	6.725	9	0.60	36.4
3	B23	7.115	2	0.96	13.67
4	Checqured Plate	Area = 7.56 × 6.725 = 50.841 m^2		W_q = 24 KN	

5. Weight of AHU platform = 3 × 1550 kg

3 No of AHU

= 4653 kg = $\frac{4653 \times 9.81}{1000}$

= 45.6 KN.

Including 15% for other minor Arrangent and Connection

DL = (45.6 + 23.14 + 36.4 + 13.67 + 24) 1.15

= 164.23 ≙ 165 KN.

Neglecting LL at the line of Earthquake

Factored Dead Load at platform level

= 1.4 × 165 = 231 KN.

Assume horizontal earthquake Coefficient 0.1 (Conservative)

∴ Net Lateral Force = 0.1 × 231

= 23.1 KN.

Assume load to be transferred through only one Column.

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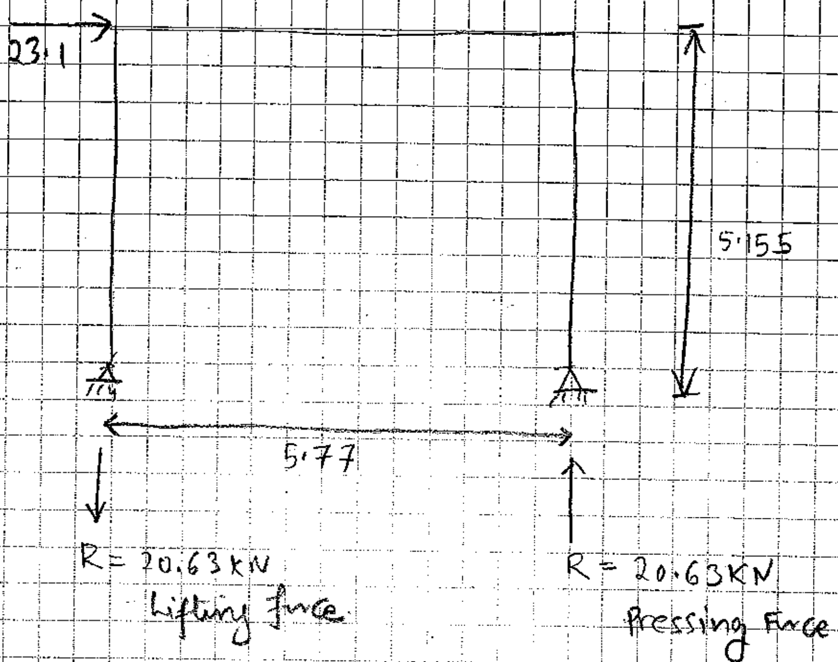
Drawing ref.	Calc. by Abhijeet / Date Apr' 2009	Check by	Date

Ref.	Calculations	Output
		

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Project	DUBAI METRO			Job ref.	
Part of Structure	AHU SUPPORT PLATFORM			Calc. sheet no. 8 / /	rev.
Drawing ref.		Calc. by Abhijeet	Date Apr '2009	Check by	Date

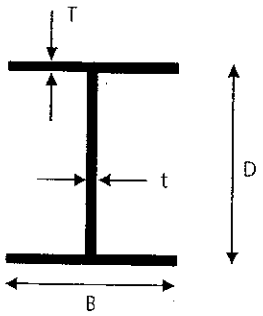
Ref.	Calculations	Output
	Column Design	
	Column Load	
	Neglected vertical Load	
	Axial Load = 333 + 101.55	
	= 434.55 KN	
	P_{max} = 434.55 + 20 = 454.55 KN.	
	P_{min} = 434.55 − 20 = 414.55 KN.	

Project:	Dubai Metro, UAE - Rashidiya , Building 3			ATKINS
Structure:	AHU Support Platform			
Calculations:	Design For Beam B1			
By	Abhijeet	**Date**	Apr'2009	**Sheet**
Ref:	**Calculations**			**Output**

BS 5950 - 1 (2000)

Steel Beam Analysis - Built Up Section

Section Properties



p_y	275 N/mm²			
D	250			
B	150			
t	6			
T	8			
d	234			
b/T	9.38	<	10ϵ	
d/t	39	<	80ϵ	*Class 2 Compact section*

Area	3804 mm²	0.298614	29.8614	
y	125 mm			
I_{xx}	4.16E+07 mm⁴			
I_{yy}	4.50E+06 mm⁴			
Z_{xx}	3.32E+05 mm³		S_{xx}	3.73E+05 mm³
Z_{yy}	5.99E+04 mm³		S_{yy}	9.42E+04 mm³
r_x	105 mm			
r_y	34 mm			

4.2.5 **Bending**

Design Moment, Mx: 25.08 kNm

Mcx = $p_y \times Z_{xx}$

91.43 kNm > Mx

Therefore; **PASS**

4.2.3 **Shear**

Design Shear Force, V:	26.5 kN				
Av	1404				
Pv	231.66 kN	>	V	Therefore;	**PASS**
0.6 x Pv	138.996 kN	>	V		**Low Shear**

Buckling

Ref			
	L	3790 m	
Table 13	L_E	5306	1.4L
	λ_{LO}	154.34	
4.3.6.8	u	1	
	x	31.3	
	λ/x	4.93895	
Table 19	v	0.77	
	β_w	1.00	

$\lambda_{LT} = uv\lambda\sqrt{\beta_w}$ = 118.8

As $\lambda_{LT} < \lambda L0$ hence there is no need to check fot LTB.

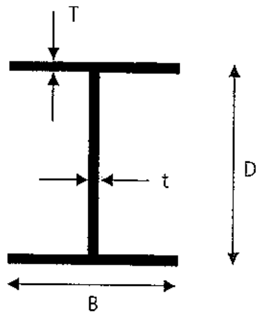
Project:	Dubai Metro, UAE - Rashidiya , Building 3			ATKINS
Structure:	AHU Support Platform			
Calculations:	Design For Beam B4			
By	Abhijeet	Date	Apr'2009	Sheet

Ref:	Calculations	Output

BS 5950 - 1 (2000)

Steel Beam Analysis - Built Up Section

Section Properties



p_y	275 N/mm²		
D	300		
B	250		
t	6		
T	12		
d	276		
b/T	10.42	<	15ϵ
d/t	46	<	80ϵ

Class 3 semi compact

Area	7656 mm²		
y	150 mm		
I_{xx}	1.35E+08 mm⁴		
I_{yy}	3.12E+07 mm⁴		
Z_{xx}	9.00E+05 mm³	S_{xx}	9.78E+05 mm³
Z_{yy}	2.50E+05 mm³	S_{yy}	3.80E+05 mm³
r_x	133 mm		
r_y	64 mm		

4.2.5 **Bending**

Design Moment, Mx: 82 kNm

Mcx = py x Z_{xx} — Taken Elastic Section modulus (Conservative)

247.50 kNm > Mx

Therefore; **PASS**

4.2.3 **Shear**

Design Shear Force, V: 49 kN

Av	1656			
Pv	273.24 kN	> V	Therefore;	**PASS**
0.6 x Pv	163.944 kN	> V		**Low Shear**

Buckling

Ref			
	L	6725	
Table 13	L_E	5043.75	0.75Lt
	λ_{LO}	78.95	
4.3.6.8	u	1	
	x	25.00	
	λ/x	3.16	
Table 19	v	0.9	
	β_W	1.00	

$\lambda_{LT} = uv\lambda\sqrt{\beta_W}$ = 71.1

As $\lambda_{LT} < \lambda LO$ hence there is no need to check fot LTB.

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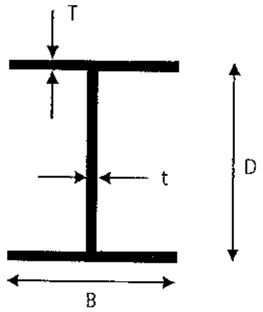
Project:	Dubai Metro, UAE - / I Rashidiya , Building 3			ATKINS
Structure:	AHU Support Platform			
Calculations:	Design For Beam B23			
By	Abhijeet	Date	Apr'2009	Sheet

Ref:	Calculations	Output

BS 5950 - 1 (2000)

Steel Beam Analysis - Built Up Section

Section Properties



p_y	275 N/mm²			
D	500			
B	250			
t	10			
T	15			
d	470			
b/T	8.33	<	9ϵ	
d/t	47	<	80ϵ	***Plastic section***

Area	12200 mm²		
y	250 mm		
I_{xx}	5.28E+08 mm⁴		
I_{yy}	3.90E+07 mm⁴		
Z_{xx}	2.11E+06 mm³	S_{xx}	2.37E+06 mm³
Z_{yy}	3.12E+05 mm³	S_{yy}	4.92E+05 mm³
r_x	208 mm		
r_y	57 mm		

4.2.5 **Bending**

Design Moment, Mx: 473 kNm

Mcx = py x Z_{xx} — Taken Elastic Section modulus (Conservative)

652.03 kNm > Mx

Therefore; **PASS**

4.2.3 **Shear**

Design Shear Force, V: 333 kN

Av	4700			
Pv	775.5 kN	>	V	Therefore; **PASS**
0.6 x Pv	465.3 kN	>	V	**Low Shear**

Buckling

Ref			
	L	5770	
Table 13	L_E	4327.5	0.75Lt
	λ_{LO}	76.52	
4.3.6.8	u	1	
	x	33.33	
	λ/x	2.30	
Table 19	v	0.96	
	β_w	1.00	

$\longrightarrow \lambda_{LT} = uv\lambda\sqrt{\beta_w} = 73.5$

As $\lambda_{LT} < \lambda L0$ hence there is no need to check fot LTB.

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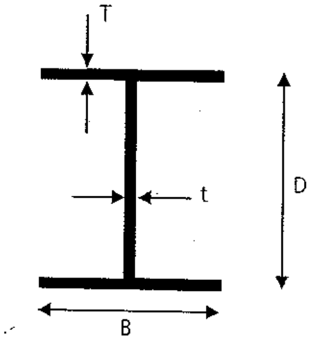
Project:	Dubai Metro, UAE - , Rashidiya , Building :3		**ATKINS**
Structure:	AHU Support Platform		
Calculations:	Design For Column C		
By	Abhijeet	**Date** APR'2009	**Sheet**

Ref:	**Calculations**	**Output**

BS 5950 - 1 (2000)

Column Analysis - Built Up Section

Section Properties



p_y	275 N/mm²			
D	300 mm			
B	300 mm			
t	10 mm			
T	15 mm			
d	270 mm			
b/T	10	<	9ε	
d/t	27	<	80ε	*Class 1 plastic section*

Area	7200 mm²		
y	150 mm		
I_{xx}	1.99E+08 mm⁴		
I_{yy}	6.75E+07 mm⁴		
Z_{xx}	1.33E+06 mm³	S_{xx}	1.46E+06 mm³
Z_{yy}	4.50E+05 mm³	S_{yy}	6.89E+05 mm³
r_x	166 mm		
r_y	97 mm		

4.7.4 **Pc - Compression Resistance**

Pc = Aq x pc

3.4.1 Aq = 7200 mm²

4.7.5 pc - From Strut curve specified in Table 23

Ref				
	L =	5.15	mm	
	L_E =	5.15	mm	(Conservatively)
	λ =	53.2		
Table 23	pc =	287	N/mm²	(for welded sections, deduct 20N/mm²)

Pc = 2066.4 kN

Project:	Dubai Metro, UAE - AL RASH-Building 3		ATKINS
Structure:	AHU Support Platform		
Calculations:	Design For Column C1		
By	Abhijeet	Date AR 2009	Sheet A 2

Ref:	Calculations	Output

Mbs & My - Compression Resistance

Mb = Pb x Zxx

4.3.6.8

u	1	
x	20	
x/λ	0.38	$\longrightarrow \lambda_{LT} = uv\lambda\sqrt{\beta_w}$ = 53.2
v	1	
β_w	1.00	

Table 19

Table 17

pb = 151 N/mm²

m_{LT} = 1

M_{bs} = 200.66 kNm

> Mx Therefore; **PASS**

M_{cy} = Py x Zyy

M_{cy} = 67.93 kNm

Design Loads

Fc = 414 kN

Mx = 0 kN-m

My = 0 kN-m

4.7.7 **Columns in Simple Structures**

Check

$$\frac{Fc}{Pc} + \frac{Mx}{Mbs} + \frac{My}{Mcy} \leq 1$$

$$\frac{414}{2066.4} + \frac{0}{200.66} + \frac{0}{67.93} = 0.20 \leq 1$$

PASS

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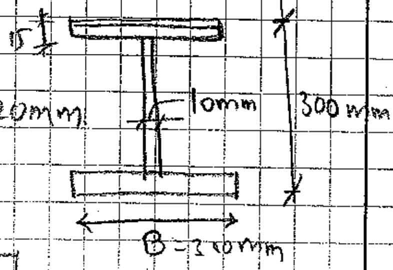
Ref.	Calculations	Output

Design of Base Plate

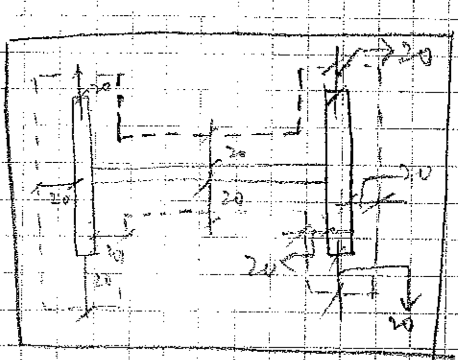
Base Plate Dimension 350x350

Column Dimension



t_p thickness of base plate = 20mm

P_{yp} = 275 N/mm²



C – Spread Width of Load
Assume spread takes place at 45°

$C = t_p = 20$

A_e Effective Bearing area

$A_e = (50 \times 340) \times 2 + 280 \times 50$
$= 34000 + 14000$
$= 48000\ mm^2$

$P_c = 454.5\ KN$

$\sigma = W = \frac{454.5 \times 1000}{A_e} = \frac{454.5 \times 1000}{48000} = 9.46\ \frac{N}{mm^2}$

$\sigma_{permissible} = 0.6 f_{cu}$

$f_{cu} = 40\ MPa$

$\therefore \sigma_{permissible} = 0.6 \times 40 = 24\ N/mm^2 > 9.46$ ok

$t_p = C\left[\frac{3W}{P_{yp}}\right]^{0.5} = 20 \times \left[\frac{3 \times 9.46}{275}\right]^{0.5} = 6.42mm < 20\ mm$ OK.

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Anchor Bolt Design :-

Total Horizontal Shear = 23.1 KN (Force on 4 bolts)

Shear on each bolt = $\frac{23.1}{4}$ = 5.775 KN

Net Tensile Force on each bolt = 0 KN.

Bolt specification
Concrete anchor bolt to be HIT-HY 150 injection mortar with HAS ROD M16 x125/38

Tensile Capacity = 15.3 KN
Shear Capacity = 24.7 KN

Check for combined Shear and Tension

$$\frac{0}{15.3} + \frac{5.775}{24.7} = 0.234 < 1.0$$ Hence OK

HIT-HY 150 injection mortar with HAS rod

Recommended load, L_{rec} [kN]: concrete $f_{ck,cube} = 25\ N/mm^2$

Anchor size	M8	M10	M12	M16	M20	M24
Tensile, N_{Rec}	6.0	8.0	12.0	15.3	26.0	32.4
Shear, V_{Rec}	5.6	9.0	13.1	24.7	38.6	55.6

Basic loading data (for a single anchor): HIT-HY 150 with HAS-R, HAS-E-R, HAS-HCR

All data on this page applies to

- concrete: See table below.
- correct setting (See setting operations page 260)
- no edge distance and spacing influence
- steel failure

For detailed design method, see pages 261 – 265.

CONC non-cracked concrete

Mean ultimate resistance, $R_{u,m}$ [kN]: concrete ≅ C20/25

Anchor size	M8	M10	M12	M16	M20	M24
Tensile, $N_{Ru,m}$	24.8	39.6	57.8	109.1	170.3	244.4
Shear, $V_{Ru,m}$	14.8	23.8	34.5	65.4	102.1	146.9

Characteristic resistance, R_k [kN]: concrete ≅ C20/25

Anchor size	M8	M10	M12	M16	M20	M24
Tensile, N_{Rk}	23.0	36.7	53.5	101.0	157.6	226.3
Shear, V_{Rk}	13.7	22.0	32.0	60.5	94.5	136.0

Following values according to the

Concrete Capacity Method

Design resistance, R_d [kN]: concrete $f_{ck,cube} = 25\ N/mm^2$

Anchor size	M8	M10	M12	M16	M20	M24
Tensile, N_{Rd}	8.4	11.2	16.8	21.4	36.4	45.4
Shear, V_{Rd}	8.8	14.1	20.5	38.8	60.6	87.2

Recommended load, L_{rec} [kN]: concrete $f_{ck,cube} = 25\ N/mm^2$

Anchor size	M8	M10	M12	M16	M20	M24
Tensile, N_{Rec}	6.0	8.0	12.0	15.3	26.0	32.4
Shear, V_{Rec}	6.3	10.1	14.6	27.7	43.3	62.3

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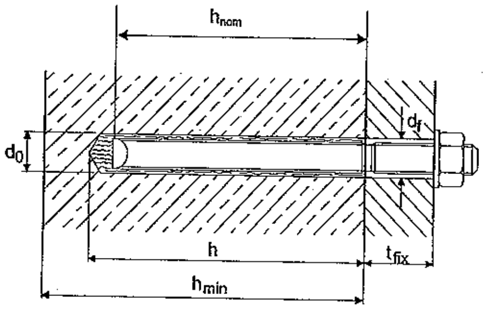
Setting details



Anchor size	M8	M10	M12	M16	M20	M24
Anchor rod [1) HAS /~~E/-R/-E-R/-HCR~~	M8x80/14	M10x90/21	M12x110/28	M16x125/38	M20x170//48	M24x210/54
d_0 [mm] Drill bit diameter	10	12	14	18	22	28
h [mm] Hole depth	80	90	110	130	175	215
h_{nom} [mm] Nominal anchorage depth	80	90	110	125	170	210
h_{min} [mm] Min. base material thickness	110	120	140	170	220	270
t_{fix} [mm] Max. fixture (fastenable) thickness	14	21	28	38	48	54
d_f [mm] Clearance hole rec.	9	12	14	18	22	26
d_f [mm] Clearance hole max.	11	13	15	19	25	29
T_{inst} [Nm] Tightening torque HAS/-E	15	30	50	100	160	240
T_{inst} [Nm] Tightening torque HAS-R/-E-R, HAS-HCR	12	25	40	90	135	200
Filling Volume[2) (guide) ml	4	6	10	15	31	65
Filling Volume[2) (guide) Trigger pulls~	0.5	1	1.5	2	4	8
Drill bit TE-CX-	10/22	12/22	14/22	-	-	-
Drill bit TE-T-	-	-	-	18/32	24/32	28/52

[1) The values for the maximum fixture thickness are only valid for the HAS anchor rods given in this table. If other HAS rods are used, these values will change. (Example: HAS M12x110/128; t_{fix} = 128 mm)

[2) The hole must be filled about 50%.

Note: To ensure that optimal holding power is obtained, the first **two trigger pulls** of mortar after opening a **330 ml** foil pack of Hilti HIT-HY 150 must be thrown away, in case of using a **500 ml** foil pack the first **three trigger pulls** must be thrown away. One trigger pull is approx. 8 ml mortar when using the MD 2000.

Temperature of the base material °C	Working time in which rod can be inserted and adjusted, t_{gel}	Curing time before anchor can be fully loaded, t_{cure}
-5	90 min.	6 hours
0	45 min.	3 hours
5	25 min.	1.5 hours
20	6 min.	50 min.
30	4 min.	40 min.
40	2 min.	30 min.

The foil pack temperature must be at least +5°C.

3

Installation equipment

Rotary hammer (TE1, TE 2, TE5, TE 6, TE6A, TE15, TE15-C, TE18-M, TE 35, TE 55 or TE 76), a drill bit, the MD 2000 or BD 2000 (P3000 F, MD2500, P3500F, P5000 HY, HIT P-8000 D) dispenser, blow-out pump, a brush and a torque wrench.

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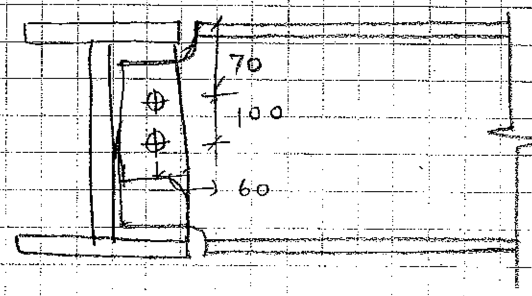
Ref.	Calculations	Output

Connection Design :-

Between B3, B1



d Nominal dia of Bolt = 20mm

φ hole dia = 22mm

F_v Shear Force = 26.47 KN

F_s Shear in Bolt

$$F_s = \sqrt{F_{sv}^2 + F_{sm}^2}$$

$$F_{sv} = \frac{F_v}{n} = \frac{26.47}{2} = 13.23 \text{ KN}$$

$$F_{sm} = \frac{F_v \times a}{Z_{bg}} \qquad a = 70 \text{ mm}$$

Z_{bg} = Elastic Section Modulus of bolt group

$$= \frac{n(n+1)p}{6} = \frac{2\times1}{6}\times100 = \frac{200}{6} = 33.33 \text{ mm}^3$$

$$F_{sm} = \frac{26.47\times70}{33.33} = 55.6 \text{ KN}$$

$$F_s = \sqrt{21.5^2+55.6^2} = 61.5 \text{ KN.}$$

(66)

Shear capacity of bolt group = 2 x 58.8 = 117.6 KN.
> 114.42 KN.

Shear capacity 16 dia bolt of Grade 8.8.

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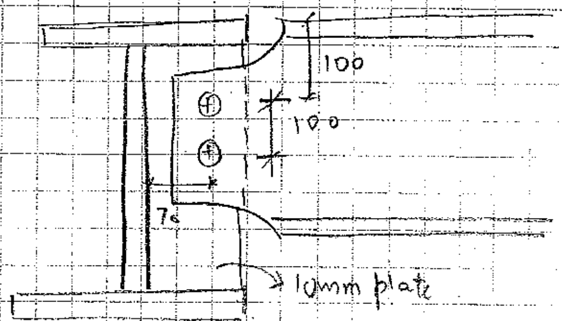
Ref.	Calculations	Output

Bearing Capacity Of Bolt 180.0 KN >

Bearing Capacity of Connected part 82.8 KN > 61.5 KN

Hence Safe.

Connection between B4 and B23



$F_V = 48.6\ KN$

$F_S = \left[(F_{SV})^2 + (F_{SM})^2\right]^{\frac{1}{2}}$

$F_{SV} = \frac{48.6}{2} = 24.3\ KN$ $\quad F_{SM} = \frac{F_V \times a}{Z_{bg}} = \frac{48.6 \times 70}{n(n+1)P}$

$= \frac{48.6 \times 70}{2 \times 100} = 17.01\ KN$

$\therefore F_S = \sqrt{24.3^2 + 17.01^2}$

$= 29.66\ KN.$

Bolt dia = 20 mm

Grade 8.8 bolt

Shear Capacity of bolt = 91.88 KN

Number of bolt = 2

Shear Capacity = 91.88 × 2 = 183.76 KN > 29.66 KN Safe.

Bearing capacity of bolt = $K_{bs} \times d \times t \times P_{bs}$

$= 1.0 \times 22 \times 10 \times 1000 = 220\ KN > 29.66$

Bearing Capacity of connected part $= K_{bs} \times d \times t \times P_{bs}$

$= 1.0 \times 22 \times 10 \times 460 = 101.2\ KN$

> 29.66 OK.

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	Part of Structure AHU SUPPORT PLAT FORM			Calc. sheet no. 18 /	rev. /
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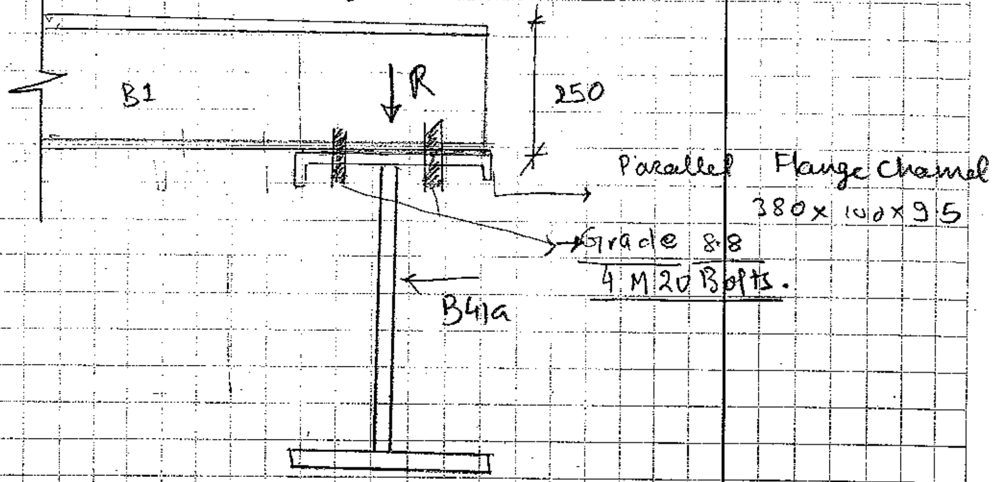
Ref.	Calculations	Output

Connection between Column and Beam:-

→ 350x10 Bearing plate at Top of column is adequate in bearing pressure

→ Connection between bearing plate and column should be rigid enough with fillet weld of size 8mm.

Connection between B1 and B41a.



R = 26.47 KN

Bearing area = 380 x 150 = 57000 mm^2

Net Bearing Pressure = $\frac{26.47 \times 1000}{57000}$ = 0.46 N/mm^2 << 460

Hence Adequate.

(6B)

ATKINS	Project : RASHIDIYA BUILDING 3			Job ref. DUBAI METRO	
WS Atkins & Partners Overseas	Part of structure BEAM B1 AND BEAM B4			Calc sheet no. / /	Rev.
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REFERENCE	CALCULATIONS	OUTPUT

Deflection Check

Beam - B1 Secondary Beam - Shear connection at the end

w	UDL Applied to the beam	=	14	kNm
L	Length of the beam	=	3.79	m
E	Modulus of elasticity of steel	=	2.E+05	N/mm^2
I_{xx}	In plane moment of Inertia	=	4.E+07	mm^4

δmax Maximum deflection at mid $= \dfrac{5 \times w \times L^4}{384 \times E \times I_{xx}}$

$= 4.525$ mm

δ_{per} Permissible deflection at mid $= \dfrac{L}{360}$

$= 10.53$ mm **Hence Safe**

Deflection Check

Beam - B4 Support Purlin - shear connection at the end

w	UDL Applied to the beam	=	14.45	kNm
L	Length of the beam	=	6.725	m
E	Modulus of elasticity of steel	=	2.E+05	N/mm^2
I_{xx}	In plane moment of Inertia	=	1.E+08	mm^4

δmax Maximum deflection at mid $= \dfrac{5 \times w \times L^4}{384 \times E \times I_{xx}}$

$= 14.253$ mm

δ_{per} Permissible deflection at mid $= \dfrac{L}{360}$

$= 18.68$ mm **Hence Safe**

ATKINS	Project : RASHIDIYA BUILDING B3			Job ref. DUBAI METRO
WS Atkins & Partners Overseas	Part of structure BEAM 23			Calc sheet no. / / Rev.
PO Box 5620 DUBAI - United Arab Emirates	Drawing ref. :	Calc. by ABHIJEET	Date Apr '2009	Check by / Date

REFERENCE	CALCULATIONS	OUTPUT

Deflection Check

Beam - B23 Secondary Beam - Shear connection at the end

w	UDL Applied to the beam	=	100	kNm
L	Length of the beam	=	5.77	m
E	Modulus of elasticity of steel	=	2.E+05	N/mm^2
I_{xx}	In plane moment of Inertia	=	5.E+08	mm^4

δmax Maximum deflection at mid $= \dfrac{5 \times w \times L^4}{384 \times E \times I_{xx}}$

$= 13.675$ mm

δ_{per} Permissible deflection at mid $= \dfrac{L}{360}$

$= 16.03$ mm **Hence Safe**

4. Appendix 1 (Drawing List)

The following is a list of drawings associated with this submission:

Drawing No.	Drawing Title
DM001-E-MDP-CDP-DD-DCC-370401-A13	Main depot Rashidiya general notes and typical details – Sheet 1
DM001-E-MDP-CDP-DD-DCC-370402-B1	Main depot Rashidiya general notes and typical details – Sheet 2
DM001-E-MDP-CDP-DD-DCC-370403-A6	Main depot Rashidiya general notes and typical details – Sheet 3
DM001-E-MDP-CDP-DD-DCC-370404-A1	Main depot Rashidiya general notes and typical details – Sheet 4
DM001-E-MAD-WGO-DD-DCC-370700-D1	Structural drawing list
DM001-E-MAD-WGO-DD-DCC-370701-D1	Foundation layout plan
DM001-E-MAD-WGO-DD-DCC-370706-D1	Level 2 layout plan.
DM001-E-MAD-WGO-DD-DCC-370710-A1	AHU Platform Layout Plan and Details
DM001-E-MAD-WGO-DD-DCC-370717-D1	Layout plan for gridline 3-13 to 3-18/3G
DM001-E-MAD-WGO-DD-DCC-370720-A1	Cat ladder and access platform details – Sheet 3
DM001-E-MAD-WGO-DD-DCC-370745-C1	Level 2 Slab Reinforcement detail.
DM001-E-MAD-WGO-DD-DCC-370751-B1	Equipment Exhaust Duct Support Frame Details
DM001-E-MAD-WGO-DD-DCC-370797-B1	Footing Details-Sheet

(Note: Drawings DM001-E-MDP-CDP-DD-DCC-370401-A13, DM001-E-MDP-CDP-DD-DCC-370402-B1, DM001-E-MDP-CDP-DD-DCC-370403-A6, and DM001-E-MDP-CDP-DD-DCC-370404-A1 have been issued on this submission for reference only)